Context selection is a partial order: A Blackwell framework and verified LLM evidence

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Abstract

Choosing what to put in a language model's prompt is almost always done by scoring each candidate source with a single number and keeping the top few. We argue this is the wrong approach. The best context for one task is often useless for another. Two sources can be *incomparable*, in the exact sense of Blackwell's 1953 comparison of experiments [Blackwell, 1953], and when they are, no single relevance score can rank them correctly for every task. To test this on real systems we use a simple check: run each source through a battery of pass/fail tasks, compare the pass rates, and attach confidence intervals so the comparison is auditable (a distribution-free *verification*). On six language models from four vendors we confirm ($\geq 90\%$ confidence, both directions) that two hand-built sources are incomparable. On a designed trap task a lexical score, two dense retrievers, and a cross-encoder reranker all prefer the wrong source; only a listwise LLM reranker, which reads the deciding detail, gets it right, exactly the boundary the theory predicts. We also find that adding a strictly larger source can *hurt*: pass rates fall from 100% to as low as 0%, and controls rule out word order and length as the cause. The lesson is practical: treat context selection as a partial-order problem, not a scalar ranking, and compare sources by a verified pass-rate test rather than a relevance number. A companion paper gives the selection rule this implies; scaling it to large corpora is the main open problem.

1 Introduction

Suppose a coding assistant is handed one of two short notes about a software module. The first note says how to *call* its main function: the order of the arguments, a keyword that is mandatory, and the error it raises on failure. The second note says how the module *encodes* numbers: a digit format ending in a checksum. For a task that calls the function the first note is decisive and the second is useless; for a task that encodes a number it is the reverse; and neither note can be reconstructed from the other. This everyday situation is the entire paper in miniature. It has a precise name, the two notes are *incomparable* in the sense of Blackwell's comparison of experiments [Blackwell, 1953], and a precise consequence: no single relevance score can rank the two notes so that the ranking is right for every task, because a score must crown one winner and here neither note wins outright.

The body makes this exact, gives a statistical test that *verifies* it from ordinary pass/fail runs on a live model, and finds that six production language models across four vendors all exhibit it. A reader who keeps these two notes in mind can read every symbol below as a statement about the first note versus the second (in the body they are the sources W_1 and W_2).

Production language-model systems spend much of their budget deciding *what context to place in the prompt*. Retrieval-augmented generation ranks passages, in-context learning selects demonstrations, and agentic coding tools choose which files and conventions to surface. Across all of these, the operative tool is a **scalar score**, lexical or embedding relevance, mutual information, $\mathcal{V}$-usable information [Ethayarajh et al., 2022], expected information gain, used to impose a total order and retain the top candidates. The scalar is convenient: it sorts, thresholds, and composes. We argue it is also the wrong abstraction, and that a classical apparatus from statistical decision theory says precisely why.

A folklore observation, restated as structure. Practitioners already know the phenomenon: *relevance does not transfer across tasks*. A source that helps one task hurts another, so "the best context" depends on the task, and this is usually treated as a nuisance to engineer away with better rankers. We show it is instead the signature of a *partial order*. Model a context source W as a statistical experiment on what the task actually needs, S (the thing a verifier checks; §3). Blackwell compares two experiments by asking whether one is a blurred copy of the other: we write $W_1 \succeq_B W_2$ when W_2's signal is a noised version of W_1's, and by the Blackwell–Sherman–Stein theorem this holds exactly when an ideal decision-maker does at least as well with W_1 as with W_2 on *every* task [Blackwell, 1953, Torgersen, 1991]. The catch is that the order is only *partial*: often neither source is a blurred copy of the other, so the two are *incomparable*. A single score cannot capture a partial order. A score forces every source into one line, but two incomparable sources each win on some task, so whichever way the score orders them, it is wrong on one.

What is and is not new. The mathematics is classical, the garbling order, the Blackwell–Sherman–Stein equivalence, Le Cam's deficiency; what is new is applying it to context selection, reading cross-task non-transfer as a partial order, and giving a test that checks it on a real model (§2). One gap needs care. Blackwell's theorem is about the *best possible* use of a source, but every number we measure comes from one fixed model: $\mathrm{PASS}_M(W, T) = \mathbb{E}[v_T(\delta_M(T, w))]$, the pass rate of a single rule δ_M that is not optimal. So we state the order twice, once for the ideal user and once for the fixed model, and prove a one-way bridge from what we measure to what the theorem is about (§4); and we call our estimator a *value-deficiency surrogate*, not a Le Cam deficiency (§6).

A latent-coordinate model that makes the order non-vacuous. One technical point keeps the framing from being empty. A fixed string of text looks like a constant, and a constant carries no information in Blackwell's sense. We avoid this by treating the hidden requirement as a list of separate facts and each source as revealing some of them; two sources are then incomparable exactly when each reveals a fact the other hides, which is easy to check and holds for our pair (Lemma 1).

Empirical findings. We test the theory on the two notes from the opening example: an API call contract W_1 and a wire-encoding rule W_2 for the same module (§7), graded by a small battery of tasks whose verifier returns pass or fail (exit code 0 is pass). The battery is a fixed, hand-built opposition, not a random sample of tasks, and the verified result rests on one API task and one encoding task; we say so plainly and do not call it a "distribution." On this setup, from *real* model calls with no invented numbers, we find:

- **Verified empirical incomparability on all six models.** The uniform verification clears $\geq 90\%$ confidence in *both* directions: the two-sided lower bounds $L_{\mathcal{D}}(W_1, W_2) \,/\, L_{\mathcal{D}}(W_2, W_1)$ are Haiku-4.5 $+0.76 \,/\, +0.76$, Sonnet-4.6 $+0.72 \,/\, +0.76$, Opus-4.8 $+0.72 \,/\, +0.76$, GPT-5.5 $+0.76 \,/\, +0.76$, DeepSeek-V4-Pro $+0.35 \,/\, +0.72$, and Kimi-K2.6 $+0.68 \,/\, +0.76$ (all n=40). Incomparability is therefore not a single-lab artifact.

- **Representative topical rankers mis-rank the pair.** On a trap task saturated with encoding vocabulary, query-conditioned lexical relevance scores $\varphi(W_1)$=0.36 $< \varphi(W_2)$=0.44 (ranking $W_2 \succ W_1$), yet realized PASS ranks $W_1 \gg W_2$ (e.g. 96–100% vs. 0%), and the same mis-rank is realized by two dense bi-encoders and a cross-encoder reranker, while an LLM listwise reranker ranks the trap correctly (§7.4.2). On the honest tasks every ranker agrees with PASS; failure occurs *only* where surface relevance diverges from verifier-decisive content, exactly the boundary Proposition 2 predicts: topical feature-based rankers must fail there, while rankers that read the decisive coordinate can escape.

- **Verified anti-monotonicity.** The superset $W_{1+} = W_2 \mathbin{++} W_1$ (the operator $++$ is text concatenation: W_2's note followed by W_1's), which literally *contains* W_1's signal, sharply degrades the API cells: Sonnet and Kimi collapse 100% $\to$ 0% (interference $\Psi_T = -1.00$, verified upper bound ≤ -0.69); GPT-5.5 verifies $\Psi_T \leq -0.60$ on `api_post_ok`. The interference verification clears on at least one cell for five of six models (Opus reproduces the collapse, 100% $\to$ 38%, but its milder drop and partly-guessable baseline keep it below the bar; §7.4.3). A length-matched padding control shows equal-length irrelevant text is harmless, so the harm is the added signal, not token count or position. One caveat: the effect is large and robust across the capability ladder but is *not* monotone in capability (Sonnet, the middle model, is the starkest; Opus, the strongest, less so), so we claim no capability trend.

- **A supporting observation: private-context value is model-relative.** The `none`-arm baselines span 0%–75% guessability across vendors (§7.4.4), an empirical instance of the theory's θ-dependence; we report it as supporting evidence that caps which cells can be verified per model, not as a headline claim.

How we ran the models. All six models are run the same way: a single API call per attempt, from which we extract the candidate code and grade it with the verifier. This keeps the setup uniform across vendors, so a difference between models reflects the model, not how it was run.

Contributions. The paper makes one claim, context selection is a partial-order problem rather than a scalar-ranking problem, and contributes exactly the three pieces that claim needs.

(C1) **Theory: context sources form a Blackwell partial order** (§4, §5). We formalize context sources as statistical experiments on the task's latent requirement, instantiate the Blackwell garbling order with its for-all-tasks dominance guarantee via the Blackwell–Sherman–Stein theorem, define the δ_M-restricted order the experiments probe, and prove a one-way bridging lemma between them. The central consequence is incomparability: Blackwell-incomparable sources cannot be ranked by *any* query-independent scalar $\varphi(W)$ for all tasks (Theorem 2), with the query-conditioned topical case isolated separately (Proposition 2). Information retrieval has long argued that no single fixed document score is optimal once documents interact (§2.1); we derive the impossibility as a theorem in the Blackwell frame.

(C2) Verification: a measurable deficiency surrogate with finite-sample guarantees (§6). We define $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = \max(0, \sup_{T\in\mathcal{D}}[\mathrm{PASS}_M(W_2, T) - \mathrm{PASS}_M(W_1, T)])$, a decision-restricted value-deficiency surrogate (named as such, not as a Le Cam deficiency), and a distribution-free uniform verification: exact Clopper–Pearson intervals [Clopper and Pearson, 1934] at Bonferroni level [Bonferroni, 1936] $\alpha = \eta/(2|\mathcal{D}|)$, valid jointly on a single event, so that $L_{\mathcal{D}} > 0$ verifies a positive value gap at confidence $\geq 1 - \eta$; we argue why a naive DKW bound [Dvoretzky et al., 1956] under-covers the supremum, and give the matching interference verification. This is the instrument that makes the partial order testable on a real model behind a real verifier.

(C3) Evidence: verified incomparability and scalar-ranking failure on live models (§7). Verified empirical incomparability on six models across four vendors; the ranker-spectrum mis-rank on a confound-controlled trap (lexical cosine, dense bi-encoders, and a cross-encoder all fail; an LLM listwise reranker that reads the decisive content escapes, exactly the topicality boundary the theory draws); and verified anti-monotonicity, a strict superset of a sufficient source collapsing PASS, with order, length-padding, and routing-instruction controls, all with the scope limits stated in §8.5.

Roadmap. §2 positions the work against scalar context-selection criteria and the recent transport of comparison-of-experiments into machine learning, and frames our delta as application-plus-framing-plus-estimator rather than new core mathematics. §3 gives the latent-coordinate experiment model and defines θ. §4 develops the partial order, the δ_M-restricted order, the non-refinement construction, and the bridging lemma, and §5 proves the source-level incomparability statement and isolates the query-conditioned case (C1). §6 defines the surrogate, the uniform verification, and the interference verification, and relates the surrogate to the true Le Cam deficiency (C2). §7 reports the six-model program with full per-cell tables (C3). §8.5 states threats to validity and scope, and §9 concludes. Proofs of the two central impossibility results (Theorem 2, Proposition 2) accompany their statements; the remaining proofs and technical material are deferred to Appendix A, the extended experimental record to Appendix B, and protocol and artifact details to Appendix C. Every reference has been verified against arXiv/DBLP/publisher records.

2 Related Work

Positioning. The tools we use are classical, Blackwell's garbling order [Blackwell, 1953] and Le Cam's deficiency [Le Cam, 1964, 1986]; we add no new core mathematics. Our contribution is to apply, frame, and measure. (i) We treat each candidate source as a statistical experiment on what the task needs and rank sources by the Blackwell order, which gives a guarantee that holds for every task at once, not just one (§4). (ii) We point out a direct consequence of Blackwell's theorem, that *incomparable* sources cannot be ranked by any scalar score for all tasks (§5), turning the folklore "relevance does not transfer across tasks" into a theorem and tying it to a limit information retrieval has long known for interacting documents (§2.1). (iii) We supply a value-deficiency *surrogate* with a verification that can be checked on a real model with a real verifier (§6). We know of no prior work that uses the comparison of experiments to choose a language model's context at inference time.

2.1 Scalar selection and its limits in information retrieval

Context and demonstration selection is dominated by *scalar* criteria, each inducing a total order and a top-k prefix: topical relevance, lexical or learned [Yu et al., 2024]; $\mathcal{V}$-usable information [Ethayarajh et al., 2022, Yuan et al., 2025]; expected information gain and Bayesian experimental design [Choudhury et al., 2025, Chan et al., 2025]; game-theoretic and influence-based demonstration valuation [Xie et al., 2024, Vinay et al., 2024]; and binary sufficient-context detection [Joren et al., 2024]. Each collapses the comparison of two sources to one real number and is therefore a scalar of the kind the paper rules out (Theorem 2 for query-independent scores, Proposition 2 for topical query-conditioned ones): not a competitor to out-perform but an instance of the excluded class (we prove Blackwell-monotonicity for the closest neighbour, $\mathcal{V}$-usable information, in Proposition 1, and conjecture it for the rest). Two families sit at the boundary of the class. Listwise LLM rerankers [Sun et al., 2023] are query-conditioned *set* functions, outside the theorem's scope, though they carry no dominance guarantee of any kind; and diversity-aware selection, from MMR [Carbonell and Goldstein, 1998] to DPPs [Ye et al., 2023] and submodular coverage [Huang, 2025], optimizes set objectives over embedding geometry, whose partial-order-grounded analogue (verified coverage of latent coordinates) we develop in a companion paper.

The limits of scalar relevance are one of the oldest themes in information retrieval, and we position inside that lineage rather than apart from it: the Probability Ranking Principle [Robertson, 1977] is optimal only under an independence assumption, its failure once documents interact is long documented [Fuhr, 2008, Carbonell and Goldstein, 1998], and the LLM-era version, relevance-maximizing retrieval degrading answer quality, is measured directly [LeVine and Varjavand, 2025, Qiu et al., 2026]. Our delta is the *form* of the statement: the failure becomes a theorem in the comparison-of-experiments frame, with the precise structural condition, Blackwell incomparability, under which every query-independent scalar must mis-rank, and a verified live instance on six production models.

2.2 Task-dependence and context degradation

That added context can degrade performance is by now well documented: a single irrelevant sentence tanks reasoning [Shi et al., 2023]; position matters [Liu et al., 2024]; high-scoring distractors hurt while random noise can help [Cuconasu et al., 2024]; retrieval curves are inverted-U in passage count [Jin et al., 2024]; document count hurts at fixed length [Levy et al., 2025]; added length alone hurts [Du et al., 2025]; and the effect persists across model generations [Hong et al., 2025] and agent-scaffold components [Liu, 2026]. We do not claim to discover this genus. Our anti-monotonicity finding contributes its order-theoretic form: a strict *superset* of a sufficient source, on sources separately *verified* Blackwell-incomparable, collapses PASS under a per-task interference verification with position, length, and routing controls; the algebraic object $\Psi_T < 0$ is one no monotone-submodular context-value scalar can carry. Pareto-incomparability of prompts along competing metrics *within* a single task [Zhao et al., 2025] is orthogonal to our object, incomparable *sources across tasks*.

2.3 The comparison of experiments in machine learning

The Blackwell–Le Cam apparatus has entered machine learning, but never for inference-time context. The closest work is training-*dataset* valuation: Zheng et al. [2024] rank datasets by Blackwell informativeness and show a pointwise-mutual-information *scalar* suffices in their Bayes-optimal, white-box regime, the opposite of our finding that no query-independent scalar can be faithful once the order is partial (Theorem 2). Nearby uses stop short of what we do: garbling to harden LLM evaluations [Bradley, 2024], statistical deficiency to order training *tasks* [Fosse et al., 2025], a deficiency-induced *directional* dominance in transfer learning (no incomparability, no distribution-free bound, outside language modeling) [Akdemir, 2025], comparison-of-experiments ideas in loss design [van Rooyen and Williamson, 2014], and a survey of Blackwell's theorems for AI [Paxton, 2026]. To our knowledge none states a Blackwell partial order over context sources, derives the no-scalar-ranks consequence for language models, or supplies a verified decision-restricted estimator, the three new pieces here.

3 Formalization

§2 placed the contribution against prior work; the next four sections make it precise, beginning here with the objects. We model the choice of *what context to supply* a language model as a comparison of statistical experiments on the task's latent requirement. This section fixes the objects and, crucially, separates two decision rules that the informal literature silently conflates: the *Bayes-optimal* rule, against which the classical Blackwell order is defined, and the *fixed model* rule, which is the object every empirical quantity in this paper actually measures. Keeping these distinct is what makes the bridge from theorem to experiment rigorous.

3.1 Tasks, latent requirements, and context sources

A *task* T fixes a latent requirement S_T, the object a deterministic verifier v_T checks (e.g. the intended program behaviour). We take S_T to be a *vector of coordinates*

$$S \;=\; (S_1, S_2, \ldots, S_m) \;\in\; \mathcal{S} \;=\; \mathcal{S}_1 \times \cdots \times \mathcal{S}_m, \tag{1}$$

where each coordinate S_j encodes one logically independent requirement (in our instantiation, S_1 is the private call-contract convention and S_2 is the private wire-encoding convention; see §7). Modelling S as a product of coordinates, rather than an opaque atom, is the technical device that makes the experiment model well posed for fixed-string sources; we return to this in §3.2 and Lemma 1.

Definition 1 (Context source as an experiment)**.** A *context source* W is a Markov kernel $\kappa_W : \mathcal{S} \to \Delta(\mathcal{W})$, i.e. a statistical experiment indexed by the latent parameter S, emitting a signal $w \sim \kappa_W(\cdot \mid S)$ in a signal alphabet $\mathcal{W}$ (the source's text). The experiment is identified with its kernel; the sample space is $\mathcal{W}$ and the observed datum is the supplied text w.

Definition 2 (Decision rules and PASS)**.** A *decision rule* is a (possibly randomized) map $\delta : (T, w) \mapsto \widehat{S}$ producing a candidate solution. Utility is the binary verifier $u(\widehat{S}, S) = v_T(\widehat{S}) \in \{0, 1\}$. Two decision rules matter:

(i) the *model rule* δ_M realized by a fixed language model M at a fixed decoding configuration θ (model weights and sampling parameters; see §3.3). Its realized pass rate on task T under source W is

$$\mathrm{PASS}_M(W, T) \;:=\; \mathbb{E}\big[\, v_T(\delta_M(T, w)) \,\big], \qquad w \sim \kappa_W(\cdot \mid S_T); \tag{2}$$

(ii) the *Bayes rule* $\delta^\star_T$, the (utility-optimal) decision rule for task T given a prior over S, achieving the optimal value $V^\star(W, T) := \sup_\delta \mathbb{E}[v_T(\delta(T, w))]$.

The distinction in Definition 2 is load-bearing. Blackwell's theorem (§4) is a statement about $V^\star$ (the optimal user of an experiment); every number we report, PASS_M, the deficiency estimator of §6, the interference statistic of §6.3, and the entire six-model program of §7, is a statement about the single fixed rule δ_M. We therefore develop the order in two registers and prove an explicit bridge between them (Lemma 2).

3.2 Sources as deterministic projection experiments

In our instantiation each context source is a *fixed string*: a snippet stating the private call contract (W_1), or the private wire-encoding invariant (W_2). A naive reading treats $\kappa_W(\cdot \mid S)$ as a constant kernel δ_{w_0} independent of S. That reading is degenerate: a constant kernel is Blackwell-equivalent to the null experiment, carries zero information, and induces the trivial σ-algebra, which would make any garbling argument vacuous. We avoid this by modelling each fixed source as a *deterministic experiment that observes a projection of the coordinate vector* (1).

Definition 3 (Projection sources)**.** A source W that reveals the coordinates indexed by a set $A \subseteq \{1, \ldots, m\}$ is the deterministic kernel

$$\kappa_W(\cdot \mid S) \;=\; \delta_{\pi_A(S)}, \qquad \pi_A(S) := (S_j)_{j \in A}, \tag{3}$$

i.e. supplying W lets a decision-maker condition on $\pi_A(S)$ and on nothing outside A. Write $\mathcal{F}_W := \sigma(\pi_A)$ for the sub-σ-algebra of $\mathcal{S}$ generated by the projection. In our instantiation W_1 reveals coordinate S_1 (the call contract, $A_1 = \{1\}$), W_2 reveals S_2 (the wire invariant, $A_2 = \{2\}$), and the superset $W_{1+} = W_2 \mathbin{++} W_1$ reveals both ($A_{1+} = \{1, 2\}$).

Under Definition 3 a fixed string is *not* an uninformative constant: it is a deterministic experiment whose informativeness is exactly the resolving power of its σ-algebra $\mathcal{F}_W$. The signal alphabet is the range of the projection, and the text "varies with the task" precisely because the realized coordinate values $\pi_A(S_T)$ differ across task instances T. This is the well-posed reading that makes the comparison of §4 a comparison of experiments rather than a metaphor, and it is what a rigorous treatment of deterministic sources requires [Torgersen, 1991].

3.3 The conditioning parameter θ

The model rule δ_M carries an implicit parameter θ: the model's weights and decoding configuration, equivalently its *prior over conventions* acquired in pretraining. We write $I(S; W \mid \theta)$ for the model-relative value of a private source, the information W supplies *beyond what θ already encodes*. When θ already "knows" a convention (the model can guess it without the source), W's marginal value collapses; when the convention is genuinely unknown to θ, W is maximally valuable. This θ-dependence is not a nuisance: it is the formal home of the *model-relative value of private context* measured in §7, where the guessability of a fixed convention spans 0% to 75% across vendors. Throughout, all realized-PASS quantities are implicitly indexed by θ; we suppress it except where the model-relativity is the point.

4 The Blackwell order over context sources

With sources modeled as experiments (§3), the classical comparison of experiments applies to them directly. We state the order, record that it is partial (the regime our experiments target), and, because those experiments measure a fixed model rather than the idealized decision-maker the classical theorem assumes, build an explicit bridge between its two registers.

4.1 The order and its two registers

Definition 4 (Garbling order)**.** $W_1 \succeq_B W_2$ (W_1 *Blackwell-dominates* W_2) iff there is a Markov kernel (garbling) $G : \mathcal{W}_1 \to \Delta(\mathcal{W}_2)$ with $\kappa_{W_2} = G \circ \kappa_{W_1}$; i.e. W_2's signal is a noised, post-processed version of W_1's, computable from W_1's signal alone without further access to S.

The classical content of the order is the Blackwell–Sherman–Stein theorem, which we state with hypotheses for a self-contained treatment.

Theorem 1 (Blackwell–Sherman–Stein; universal dominance for the optimal rule)**.** *Let W_1, W_2 be experiments on a common parameter space $\mathcal{S}$ with standard Borel signal spaces. The following are equivalent:*

(i) $W_1 \succeq_B W_2$ *(Definition 4);*

(ii) *for every prior μ on $\mathcal{S}$, every finite decision space, and every bounded loss ℓ, the Bayes risk under W_1 is no larger than under W_2; equivalently the optimal value satisfies $V^\star(W_1, T) \geq V^\star(W_2, T)$ for every task T with bounded utility.*

Randomized decision rules are admitted on both sides; if the dominated experiment is realized with a randomized garbling, no loss of generality is incurred by allowing randomized rules in (ii)*.*

This is the classical Blackwell–Sherman–Stein theorem [Blackwell, 1953, Torgersen, 1991], which we use as a black box. The only specialization is reading "decision problem" as a task with bounded verifier utility $v_T \in \{0, 1\}$ and "experiment" as a context source (Definition 1), which is faithful because the kernels of Definition 3 are standard Borel.

Corollary 1 (Universal context dominance, C1)**.** *If $W_1 \succeq_B W_2$ then, for the* optimal *user, supplying W_1 is at least as good as supplying W_2 on* every *task, prior, and bounded utility, a decision-independent, for-all-tasks guarantee that no task-specific scalar score (relevance, mutual information, $\mathcal{V}$-usable information) provides.*

Remark 1 (The order is partial; no lattice). $\succeq_B$ is a genuine partial order: there exist pairs W_1, W_2 with neither $W_1 \succeq_B W_2$ nor $W_2 \succeq_B W_1$, and the order does not form a lattice [Torgersen, 1991]. Incomparability is exactly the regime our experiments target.

4.2 Deterministic sources: incomparability via σ-algebra non-refinement

For deterministic projection sources (Definition 3) the garbling order reduces to refinement of the generated σ-algebras, which makes incomparability checkable by a purely structural argument rather than by an empirical divergence.

Lemma 1 (Refinement characterization for projection sources)**.** *Let W_A, W_B be projection sources revealing coordinate sets A, B (Definition 3), with coordinates $S_1, \ldots, S_m$ mutually independent under some full-support prior and each S_j non-degenerate. Then:*

(a) $W_A \succeq_B W_B \iff \mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A} \iff B \subseteq A$*;*

(b) *consequently W_A and W_B are Blackwell-*incomparable *iff $A \not\subseteq B$ and $B \not\subseteq A$, i.e. each reveals at least one coordinate the other hides.*

Lemma 1 is the formal version of the garbling argument used in §7: from the wire format (W_2, revealing S_2) one cannot manufacture the private arg-order / required-keyword / exception-type facts (S_1), and from the call contract (W_1, revealing S_1) one cannot manufacture the cents-scale / check-digit facts (S_2). With $A_1 = \{1\}$, $A_2 = \{2\}$ neither set contains the other, so W_1, W_2 are Blackwell-incomparable by construction (and W_{1+}, revealing $\{1, 2\}$, dominates both). This guarantees the *theoretical* incomparability the experiments then exhibit empirically.

4.3 Bridging the optimal-rule order to the fixed model

Theorem 1 concerns $V^\star$. Our experiments measure PASS_M for a single fixed, generally sub-optimal rule δ_M. The two are not interchangeable: an inequality between optimal values need not hold for a fixed rule, and, as we show empirically, the fixed rule can even *violate* monotonicity that the optimal value obeys. We therefore introduce a δ_M-restricted order and state precisely what it does and does not inherit from Theorem 1.

Definition 5 (δ_M-restricted, $\mathcal{D}$-restricted order)**.** Fix the model rule δ_M and a finite task set $\mathcal{D}$. Say $W_1 \succeq_{M,\mathcal{D}} W_2$ iff $\mathrm{PASS}_M(W_1, T) \geq \mathrm{PASS}_M(W_2, T)$ for every $T \in \mathcal{D}$. Call W_1, W_2 $(M, \mathcal{D})$-*incomparable* iff $W_1 \not\succeq_{M,\mathcal{D}} W_2$ and $W_2 \not\succeq_{M,\mathcal{D}} W_1$, i.e. each strictly out-performs the other on some task in $\mathcal{D}$.

$\succeq_{M,\mathcal{D}}$ is the order our verification measures. It is decision-restricted (one fixed rule) and distribution-restricted (a fixed finite $\mathcal{D}$); within those restrictions it is still a genuine, *for-all-tasks-in-$\mathcal{D}$* dominance statement, not a scalar.

Lemma 2 (One-way bridge under coordinate separation)**.** *Neither order implies the other unconditionally; each direction of the bridge carries an explicit hypothesis.*

(a) *Let W_1, W_2 be projection sources revealing coordinate sets A_1, A_2 (Definition 3), and let $T_a, T_b \in \mathcal{D}$ be tasks whose verifier-decisive coordinates lie in $A_1 \setminus A_2$ and $A_2 \setminus A_1$ respectively, with* guessing caps *$g_a := V^\star(W_2, T_a)$ and $g_b := V^\star(W_1, T_b)$, the optimal values achievable* without *the decisive coordinate. If* $\mathrm{PASS}_M(W_1, T_a) > g_a$ *and* $\mathrm{PASS}_M(W_2, T_b) > g_b$*, then W_1, W_2 are Blackwell-incomparable.*

(b) *The converse fails in general: Blackwell incomparability concerns $V^\star$, and a fixed sub-optimal δ_M may fail to realize the optimal-value crossover on $\mathcal{D}$. It holds under the value-attainment hypothesis (Assumption 1).*

The separation hypothesis in (a) *is necessary, and our own measurements witness the necessity: (W_1, W_{1+}) is $(M, \mathcal{D})$-incomparable on the measured cells (W_1 wins the API cells, W_{1+} the encoding cell) yet Blackwell*-comparable *by construction ($W_{1+} \succeq_B W_1$), precisely because the deprived-side premise fails there, W_{1+} contains W_1's coordinate, so the API-side cap is not small. A bare* PASS_M *crossover, without the separation hypothesis, witnesses only the sub-optimality of δ_M (Theorem 3).*

Intuitively: a fixed-rule crossover proves structural incomparability only when each winning source reveals a coordinate the losing source lacks; otherwise the crossover may witness nothing more than the rule's own incoherence.

Assumption 1 (Value attainment on the witnessing tasks)**.** There exist tasks T_a, T_b and a margin $\gamma > 0$ such that δ_M attains $\mathrm{PASS}_M(W_1, T_a) - \mathrm{PASS}_M(W_2, T_a) \geq \gamma$ and $\mathrm{PASS}_M(W_2, T_b) - \mathrm{PASS}_M(W_1, T_b) \geq \gamma$; i.e. the fixed model actually exploits each source's private coordinate on the task that requires it. Under Assumption 1 with $\{T_a, T_b\} \subseteq \mathcal{D}$, W_1 and W_2 are $(M, \mathcal{D})$-incomparable.

Lemma 2 is the keystone. In the direction the empirical program needs, it says: a verified $(M, \mathcal{D})$-crossover on tasks whose decisive coordinates are *separated* across the two sources, with the deprived side's optimal value capped below the measured pass rates, implies genuine Blackwell incomparability. Our construction is designed to satisfy the hypothesis (each cell's decisive private coordinate is revealed by exactly one source, Lemma 1), and its observable signature, deprived-arm pass rates flooring at the guessing cap (W_2 at 0% on the API cells, W_1 at 0% on the encoding cell, `none` near 0), is verified per cell in §7; where a model's prior partially recovers a convention (the

θ-dependence of §3.3), the cap premise weakens and we do not verify on those cells. The reverse implication, predicting the realized crossover from the abstract theorem, requires Assumption 1, which our task design enforces and our measurements verify.

5 The incomparability theorem

§4 showed the order is partial; we now establish the consequence that motivates the whole paper: no scalar can rank a Blackwell-incomparable pair. We *split* the claim into a clean theorem for query-*independent* source scores $\varphi(W)$ and a separate, explicitly empirical statement for query-*conditioned* scores $\varphi(W, T)$ (the relevance score practitioners actually use); the two must not be conflated.

5.1 Query-independent scores: the theorem

Theorem 2 (No query-independent scalar ranks incomparable sources, C1)**.** *Let W_1, W_2 be Blackwell-incomparable. Let $\varphi : \{sources\} \to \mathbb{R}$ be any* query-independent *scalar score (a single number per source: relevance to a fixed corpus, mutual information $I(S; W)$, $\mathcal{V}$-usable information, perplexity-gap, expected information gain under a fixed prior). Then there exist tasks T_a, T_b such that the ranking of $\{W_1, W_2\}$ induced by φ disagrees with the optimal-value ranking on at least one of T_a, T_b; and, under Assumption 1, with the realized-*PASS_M *ranking on at least one of them.*

Proof. By incomparability, neither source garbles the other. By the contrapositive of Theorem 1, “not $W_1 \succeq_B W_2$” yields a task T_b (a prior plus bounded utility) with $V^\star(W_2, T_b) > V^\star(W_1, T_b)$, and “not $W_2 \succeq_B W_1$” yields a task T_a with $V^\star(W_1, T_a) > V^\star(W_2, T_a)$; both inequalities are *strict*, so ties do not arise. A query-independent φ assigns the single ordered pair $(\varphi(W_1), \varphi(W_2))$, fixing one total order of $\{W_1, W_2\}$ *independent of the task.* That fixed order agrees with the optimal-value order on at most one of T_a, T_b (they point opposite ways), hence is violated on the other. The PASS_M conclusion follows by replacing $V^\star$ with PASS_M via Assumption 1, which guarantees the realized rule reproduces both strict inequalities with margin γ. □

Remark 2 (Scope). Theorem 2 rules out exactly the class of *source-level* scalars: any φ that is a fixed number per source, regardless of task. $\mathcal{V}$-usable information, mutual information, perplexity-gap, and (fixed-prior) EIG all live in this class, so they are corollary victims of the theorem, not competitors (Proposition 1). The theorem is mathematically a one-line consequence of Blackwell's theorem; the contribution is the *application and framing*, reading cross-task non-transfer of relevance as the signature of a partial order, not new core mathematics [Blackwell, 1953, Torgersen, 1991].

Proposition 1 ($\mathcal{V}$-usable information is Blackwell-monotone)**.** *Let $\mathcal{V}$-usable information $I_{\mathcal{V}}(W \to S)$ be defined, after Ethayarajh et al. [2022], as the reduction in expected predictive Bayes risk of the best predictor in a fixed family $\mathcal{V}$ when given the signal w versus the null signal. If $W_1 \succeq_B W_2$ and the predictor family $\mathcal{V}$ is closed under pre-composition with measurable garblings, then $I_{\mathcal{V}}(W_1 \to S) \geq I_{\mathcal{V}}(W_2 \to S)$. Hence $I_{\mathcal{V}}$ is a Blackwell-monotone scalar: it respects $\succeq_B$ but collapses it to one real number.*

Remark 3 (Scope of Proposition 1). The closure hypothesis (a family closed under pre-composition with garblings) is the natural sufficient condition and holds for predictor families rich enough to apply a fixed measurable post-processing; for narrower families monotonicity may fail and we do not claim it. We prove monotonicity for the closest neighbour, $\mathcal{V}$-usable information, and conjecture, rather than prove, the analogue for mutual information, perplexity-gap, and fixed-prior EIG, each of which is a data-processing-monotone functional of the same kernels.

5.2 Query-conditioned scores: the relevance trap (empirical)

A query-conditioned score $\varphi(W,T)$, e.g. the lexical or embedding relevance of a source to the task prompt, which is what retrieval and RAG actually deploy, induces a *different* order per task and is therefore *not* a single fixed total order. Theorem 2 does *not* cover it. We make two precise statements instead.

Proposition 2 (Constrained impossibility for feature-measurable scores)**.** *Suppose $\varphi(W,T)$ is measurable with respect to a fixed feature map of the pair (W,T) that is* topical*: it depends on W,T only through surface lexical overlap and is invariant to the verifier-decisive coordinate of S_T. If there is a task T_c whose decisive coordinate is supplied by W_1 but whose surface lexicon is saturated with W_2's vocabulary, then $\varphi(W_2,T_c) > \varphi(W_1,T_c)$ while the realized utility ranks $W_1 \succ W_2$ on T_c. Hence no topical query-conditioned score ranks $\{W_1,W_2\}$ correctly on all tasks.*

Proof. By topical invariance, $\varphi(\cdot,T_c)$ ignores the verifier-decisive coordinate and orders by lexical overlap, which favours W_2 on T_c by construction; utility favours W_1 because T_c's PASS requires W_1's coordinate. The two orders disagree on T_c. $\square$

Proposition 2 is the strongest *general* statement we can make about query-conditioned scores: it holds for any score in the topical (lexical-feature) class, but not for an arbitrary learned reranker that could, in principle, read the private coordinate. We therefore treat the defeat of query-conditioned relevance as an *empirical* result, demonstrated for the concrete lexical-cosine score practitioners use. In §7 the `trap_store_wire` task realizes T_c: bag-of-words cosine gives $\varphi(W_1) = 0.36 < \varphi(W_2) = 0.44$ (ranking $W_2 \succ W_1$), yet realized PASS ranks W_1 at 96%–100% versus W_2 at 0–10% across all six models. The same mis-rank is realized by representative open-weight learned rankers: two dense bi-encoders (BGE, E5) and a cross-encoder reranker all prefer W_2 on the trap (§7.4.2, Table 3), so the topical class is not a lexical strawman: its failure mode is shared by the ranker families retrieval systems actually deploy. We also measure the boundary from the other side: an LLM *listwise* reranker, which reads both documents and can therefore condition on the private coordinate itself, ranks the trap correctly (30/30). This is the behaviour Proposition 2 predicts, its hypothesis is topicality (invariance to the verifier-decisive coordinate), and a ranker that reads the coordinate is simply outside the impossibility class.

6 The decision-restricted deficiency surrogate and its uniform verification

The impossibility of §5 concerns the idealized user of a source; every quantity we report, by contrast, comes from a single fixed model. This section builds the instrument that connects the two: a surrogate that measures the order on the fixed rule, and a distribution-free verification that makes each comparison auditable on a live model.

6.1 The surrogate and its correct name

Exact garblings over text cannot be computed, and $\succeq_B$ is only partial, so instead we measure the fixed-rule order of Definition 5. Fix the model rule δ_M and a finite task set $\mathcal{D}$, and define the *decision-restricted value-deficiency*, the worst amount by which W_1 trails W_2 across the tasks in $\mathcal{D}$:

$$\widehat{\delta}_{\mathcal{D}}(W_1,W_2) \;:=\; \max\Big(0,\ \sup_{T\in\mathcal{D}} \left[\mathrm{PASS}_M(W_2,T) - \mathrm{PASS}_M(W_1,T)\right]\Big), \tag{4}$$

estimated per cell by $\mathrm{PASS}_M(W,T) \approx k_{W,T}/n$ from n completions. Then $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$ verifies empirical dominance $W_1 \succeq_{M,\mathcal{D}} W_2$, and mutual positivity $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \widehat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$ verifies $(M, \mathcal{D})$-incomparability.

Remark 4 (This is *not* Le Cam deficiency). The Le Cam deficiency $\delta(\mathcal{E}, \mathcal{F}) = \inf_G \sup_{\text{loss}} [R_{G\circ\mathcal{E}} - R_{\mathcal{F}}]$ *minimizes over garblings* G and takes a *supremum over a class of losses* [Le Cam, 1964, 1986, Torgersen, 1991]. The quantity in (4) performs no garbling minimization and uses a single fixed rule δ_M over a fixed finite task set; it is a one-sided value/regret gap, not a deficiency. No general inequality relates $\widehat{\delta}_{\mathcal{D}}$ to the true Le Cam deficiency for a fixed sub-optimal rule (it can be larger or smaller in either direction). We therefore name (4) the *decision-restricted value-deficiency* and use "Le Cam deficiency" only as the conceptual analogy that motivates a graded relaxation of the partial order. The one direction that *does* transfer is qualitative: $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$ realizes the for-all-tasks-in-$\mathcal{D}$ dominance of Definition 5, the measurable shadow of Corollary 1.

6.2 The uniform verification

The surrogate (4) selects the *argmax-gap* task, so a pointwise confidence statement on a single task ignores the selection. We pay the multiplicity with an exact Clopper–Pearson interval per cell and a union bound, which is distribution-free and finite-sample valid.

Proposition 3 (Uniform incomparability verification)**.** *Within each cell* (W,T) *draw* n *i.i.d. Bernoulli PASS outcomes (see Assumption 2). Let* $[L_{W,T}, U_{W,T}]$ *be the exact (two-sided) Clopper–Pearson interval [Clopper and Pearson, 1934] for* $p_{W,T} = \mathrm{PASS}_M(W,T)$ *at level* $\alpha = \eta/(2|\mathcal{D}|)$*. By the Bonferroni/union bound [Bonferroni, 1936] over the* $2|\mathcal{D}|$ *proportions* $\{p_{W_1,T}, p_{W_2,T}\}_{T\in\mathcal{D}}$*, all* $2|\mathcal{D}|$ *intervals hold simultaneously with probability at least* $1-\eta$*. On that event,*

$$L_{\mathcal{D}}(W_1, W_2) \;:=\; \max_{T\in\mathcal{D}} \left[\, L_{W_2,T} - U_{W_1,T} \,\right] \tag{5}$$

is a $(1-\eta)$ *lower confidence bound on* $\sup_{T\in\mathcal{D}}[p_{W_2,T} - p_{W_1,T}]$*. Hence* $L_{\mathcal{D}}(W_1, W_2) > 0$ ***verifies*** $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ *at confidence* $\geq 1-\eta$*. Both directions* $L_{\mathcal{D}}(W_1, W_2)$ *and* $L_{\mathcal{D}}(W_2, W_1)$ *read off the* same $2|\mathcal{D}|$ *intervals, so the joint incomparability claim* $\{\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \widehat{\delta}_{\mathcal{D}}(W_2, W_1) > 0\}$ *holds at* $\geq 1-\eta$ *with no additional correction. The matching quantity* $U_{\mathcal{D}}(W_1, W_2) := \max_{T\in\mathcal{D}} \left[\, U_{W_2,T} - L_{W_1,T} \,\right]$ *is, on the same event, a* $(1-\eta)$ upper *confidence bound on the same supremum; the dominance control uses it to verify* $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$ *up to the equivalence margin.*

A naive single-sample DKW bound under-covers this supremum, because the object is a maximum of differences of binomial means across distinct cells and the verification selects the argmax-gap cell, so the multiplicity must be paid explicitly rather than absorbed into a single-sample envelope (Appendix A); the resulting construction aligns with recent statistical guidance for black-box LLM evaluation (Appendix A).

Assumption 2 (Sampling regime)**.** The n outcomes within a cell are i.i.d. Bernoulli draws of $v_T(\delta_M(T, w))$. Every draw is a single completion (plus deterministic code extraction) from an independent call with no shared state, so the draws are manifestly independent and the binomial model applies directly; we report the decoding configuration in §7. Where draws are near-deterministic ($p \in \{0, 1\}$ stark cells), Clopper–Pearson collapses and the verification is conservative.

6.3 The interference verification (anti-monotonicity)

The deficiency surrogate also detects a phenomenon the optimal-value order forbids but a fixed rule can exhibit: supplying a *superset* source W_{1+} (revealing strictly more coordinates than W_1;

Definition 3) can *degrade* performance. Define the per-task interference

$$\Psi_T \;:=\; \mathrm{PASS}_M(W_{1+}, T) - \mathrm{PASS}_M(W_1, T) - \mathrm{PASS}_M(W_2, T) + \mathrm{PASS}_M(\texttt{none}, T), \tag{6}$$

the second-difference (interaction) of adding W_2 on top of W_1. We write the interaction as Ψ_T (rather than a symbol I that would collide with mutual information $I(S; W \mid \theta)$). A verified $\Psi_T < 0$ on a cell whose W_1 arm is at ceiling is the localized statement that "more strictly-more-informative context hurts."

Proposition 4 (Interference verification)**.** *Fix a target task $T^\star \in \mathcal{D}$ chosen* before *unblinding. With per-cell exact Clopper–Pearson endpoints* $\mathrm{CP}_{\mathrm{lo}}, \mathrm{CP}_{\mathrm{hi}}$ *at level $\alpha = \eta/4$ on the four arms $\{W_{1+}, W_1, W_2, \texttt{none}\}$ of $T^\star$,*

$$\overline{\Psi}_{T^\star} \;:=\; \mathrm{CP}_{\mathrm{hi}}(p_{W_{1+}}) - \mathrm{CP}_{\mathrm{lo}}(p_{W_1}) - \mathrm{CP}_{\mathrm{lo}}(p_{W_2}) + \mathrm{CP}_{\mathrm{hi}}(p_{\texttt{none}}) \tag{7}$$

is a $(1-\eta)$ upper confidence bound on $\Psi_{T^\star}$. Hence $\overline{\Psi}_{T^\star} \le -\tau$ verifies harmful non-monotonicity $\Psi_{T^\star} \le -\tau$ at confidence $\ge 1-\eta$. If instead the harmful cell is selected *from $|\mathcal{D}|$ candidates after unblinding, use $\alpha = \eta/(4|\mathcal{D}|)$; this is the same selection correction the verification applies to the deficiency surrogate (Remark 6), and omitting it would repeat the multiplicity error we criticize.*

Remark 5 (The leaky-baseline limitation, stated as a protocol). The upper bound (7) includes $+\mathrm{CP}_{\mathrm{hi}}(p_{\texttt{none}})$, so it can verify harm only on cells whose `none`-baseline floors near 0 (the convention is unguessable for the model under test). When the baseline is non-zero, e.g. `api_argorder` at `none` = 75% for GPT-5.5, 45% for Kimi, 15% for Opus, the bound is mechanically slack and harm is not verifiable even when the point estimate is strongly negative. We therefore adopt the explicit protocol: *verify Ψ_T only on cells whose* `none`*-baseline floors near* 0 *for the model under test*, and report, per model, which cell carried the verification and what its baseline was. This makes the cell selection a stated rule rather than an outcome-driven choice. The same model-relative guessability that defines the value of private context (§3.3) thus governs which cells can carry the interference verification.

6.4 Anti-monotonicity as information non-monotonicity of δ_M

A verified $\Psi_{T^\star} < 0$ on a ceiling-W_1 cell is more than a curiosity: it is a falsification of Blackwell superset-dominance *for the fixed model rule*, and it pinpoints exactly where the optimal-rule order (Theorem 1) and the realized rule diverge.

Theorem 3 (Information non-monotonicity of the model rule)**.** *Let W_{1+} reveal a superset of W_1's coordinates, so $W_{1+} \succeq_B W_1$ (Lemma 1). Any* Blackwell-coherent *rule, one whose value is monotone under $\succeq_B$, as the Bayes rule $\delta^\star$ is by Theorem 1, satisfies* $\mathrm{value}(W_{1+}, T) \ge \mathrm{value}(W_1, T)$ *for every task T. If a fixed model rule δ_M has a task $T^\star$ with* $\mathrm{PASS}_M(W_{1+}, T^\star) < \mathrm{PASS}_M(W_1, T^\star)$*, then δ_M is* not *Blackwell-coherent: it violates monotonicity under the garbling order on $T^\star$. In particular no monotone-submodular scalar context-value score can predict δ_M's behaviour on $T^\star$.*

Theorem 3 converts the empirical collapse into a structural statement: the harm is not noise but a witness that real LLM decision rules are not Blackwell-coherent, which is precisely the gap Lemma 2(b) warned cannot be closed without value attainment. The deficiency surrogate detects this reversal because it tracks realized PASS, whereas every monotone scalar must miss it by construction.

6.5 Family-wise error across the verified claims

Each verification above controls error *marginally* at η for its own claim (incomparability via Proposition 3; interference via Proposition 4). When several claims are read off the *same* n-sample run at the same η, incomparability (both directions, on one shared event E), one-directional dominance, and the interference cells, the conjunction over distinct claims is *not* jointly controlled at η unless the budget is split across claims. We state results as marginally verified per claim and, where a joint statement is wanted, allocate η across the C verified claims (e.g. η/C each) so the global family-wise error is $\leq \eta$. The incomparability pair is the one exception that is genuinely joint at η on a single event, as proved in Proposition 3; all other cross-claim conjunctions on a shared run are reported as marginal.

The two verifications have different effective-n requirements, which is why the consolidated record marks Opus incomparability as verified and Opus interference as reproduced but not verified; the power analysis is in Appendix A.

7 Experiments

With the order (§4), its impossibility consequence (§5), and the verification that measures it (§6) in hand, we now instantiate the theory on real language models. The design has three commitments. First, every reported number is an empirical PASS rate from live model calls verified by an executable test harness; no value is modeled, simulated, or extrapolated. Second, the two context sources are constructed to be *Blackwell-incomparable by construction* (§7.1), so the incomparability theorem (Theorem 2) has a non-vacuous instance to bite on. Third, every verified claim is reported with the exact sample size, the Clopper–Pearson endpoints, and the per-cell PASS counts from which it is derived (§7.4), so the verification is auditable rather than asserted.

7.1 Design: two incomparable context sources

We define two *private, unguessable* conventions on the *same* module `ledger`, matched on every nuisance axis (length, roughly three sentences, exactly one private convention each, same module, same task shape). Because both are on-topic for any "ledger" task, any lexical, embedding, mutual-information, or $\mathcal{V}$-usable score treats them as equally relevant candidates, the precondition for Theorem 2 to apply.

W_1**, API call contract.** `ledger.post(account, amount, *, memo)`: `account` first, `amount` second positional, `memo` a *required keyword-only* argument; returns an integer entry id ≥ 1; raises `ledger.PostError` (not `ValueError`/`KeyError`) on failure.

W_2**, wire-encoding invariant.** Amounts serialize to `<sign><minor-unit-digits>#<check>`, where check = (sum of digits) mod 10; e.g. \$12.30 $\mapsto$ `"+1230#6"` and $-$\$0.07 $\mapsto$ `"-7#7"`. Decoding rejects a wrong check digit.

W_{1+}**, literal superset.** $W_{1+} = W_2 \mathbin{+\!\!+} W_1$, the concatenation of W_2's signal with W_1's. As a deterministic source revealing both coordinates, it Blackwell-dominates both W_1 and W_2 ($W_{1+} \succeq_B W_1$ and $W_{1+} \succeq_B W_2$); it is the anti-monotonicity / dominance probe. We also use $W_{1+}^{\mathrm{rev}} = W_1 \mathbin{+\!\!+} W_2$ (W_1 first) as an order control.

Why incomparable (garbling argument). The argument is simple. From W_2's wire format you cannot recover W_1's private argument order, its required `memo` keyword, or its error type; from W_1's contract you cannot recover W_2's cents scale, its `#` separator, or its mod-10 check. Each source states a fact the other leaves out, so neither is a blurred copy of the other, and W_1, W_2 are incomparable. This is exactly the coordinate picture of §3: W_1 and W_2 reveal different, non-overlapping facts about S (Lemma 1).

Confound defense: the double crossover. The key pattern is a *double crossover*, measured against a `none` baseline (no note supplied): W_1 raises the API tasks but *not* the encoding task, while W_2 raises the encoding task but *not* the API tasks. If encoding were simply "harder," W_1 would fail to help it and W_2 would still have to raise it from the same floor. In the clean carrying cells the deprived arms floor at or near the model's guessing cap, so the two task families are equally hard on their own and any difference comes from the matching private fact, not from one family being harder; when a model prior leaks a convention (Table 5), we report that leakage explicitly and do not use those cells for claims that require a near-zero baseline.

7.2 Tasks

The full design comprises seven tasks with stub-`ledger` verifiers (return code 0 = PASS): four W_1-decisive API tasks (`api_post_ok`, `api_argorder`, `api_error_type`, `api_memo_required`), two W_2-decisive encoding tasks (`enc_amount`, `dec_amount`), and one relevance trap (`trap_store_wire`). The verifier stubs a fake `ledger` that accepts *only* the exact private contract (rejecting wrong positional order, a missing `memo=`, or the wrong exception type).

Verified battery. The verified runs use a fixed four-cell battery: `api_post_ok`, `api_argorder` (hardened so `memo` arrives through a parameter, forcing reliance on the keyword-only fact), `enc_amount`, and the trap `trap_store_wire`. We disclose this reduction explicitly: the three remaining designed cells (`api_error_type`, `api_memo_required`, `dec_amount`) are not part of the verified battery. The four-cell selection was made to retain at least one W_1-decisive, one W_2-decisive, and the trap cell while keeping the cross-model spend tractable; we report below that this is best understood as a fixed adversarial four-cell battery, not a sampled task distribution, and we state the effective support of the deficiency surrogate accordingly (§7.4.1).

The relevance trap. The trap `trap_store_wire` is the confound-proof form of Theorem 2. Its prompt is lexically saturated with encoding vocabulary, so a *query-conditioned* lexical relevance score ranks $W_2 > W_1$; but its graded requirement is the W_1 call contract (it must post an already-encoded wire string through `post`), so realized PASS ranks $W_1 > W_2$. This defeats the query-conditioned relevance score practitioners actually use, not merely a query-independent ranking. We measure relevance as bag-of-words cosine $\varphi_{\text{lex}}(W, T)$ between a source and the task prompt, computed *before* the model is run. We use lexical cosine as the *minimal* instantiation of Proposition 2, the cheapest member of the topical class; stronger learned rankers (dense bi-encoders, a cross-encoder, and an LLM listwise reranker) are tested against the same trap separately in §7.4.2, while the theorem (Theorem 2) is the general claim for query-independent scalars.

7.3 Methodology

Models and vendors. We evaluate six models across four vendors: Anthropic Haiku-4.5, Sonnet-4.6, and Opus-4.8; OpenAI GPT-5.5; DeepSeek-V4-Pro; and Moonshot Kimi-K2.6.

Harness and arms. The harness (`tokenbench/measure_blackwell.py`; released with all run logs at `https://github.com/abilous-ti/blackwell-llm-context`) is Python-standard-library only (zero external dependencies); it computes $\widehat{\delta}_{\mathcal{D}}$ in both directions, the uniform Clopper–Pearson/Bonferroni verification, the interference verification, and the lexical relevance score, and it implements the launchers for the Anthropic CLI, the Azure Responses API, and OpenAI-compatible chat completions. Each cell is run under the arms $\{\texttt{none}, W_1, W_2, W_{1+}\}$. PASS is measured over n completions per cell.

One uniform run mode. All six models are queried the same way: a single completion per attempt (the Anthropic models via `claude -p` with a one-turn budget and no tools, the others through an OpenAI-style API), from which we extract the candidate code and grade it. Because the run mode is identical across vendors, a difference between models is a fact about the model, not about how it was run. This is a deliberately conservative probe of whether a model uses supplied context: it removes the agentic-exploration pathway, so the model must use what it is given in one pass.

Verification parameters. All verifications use confidence budget $\eta = 0.10$. Incomparability uses exact Clopper–Pearson intervals at per-cell level $\alpha = \eta/(2|\mathcal{D}|)$ over the $2|\mathcal{D}|$ proportions $\{p_{W_1,T}, p_{W_2,T}\}_{T\in\mathcal{D}}$, so by the union bound all hold jointly with probability $\geq 1-\eta$ and both directions read off the same intervals. The interference verification uses $\alpha = \eta/4$ and the threshold $\tau = 0.30$, verifying $\Psi_T \leq -\tau$ via the upper bound of (7). We report each verification as marginally controlled at η; we do not claim joint family-wise control across the incomparability, dominance, and interference claims drawn from a single run (§6.5, §8.5). The intervals are degenerate at $p = 0$ and $p = 1$, which is why the stark 100%/0% cells verify incomparability at small n while the interference contrast, which mixes intermediate baselines, does not.

Sampling regime. The verification models each cell as n Bernoulli PASS draws (Assumption 2); the four arms within a cell are separate completions, and each of the n draws is an independent call with no shared state, so the draws are manifestly independent. We report the decoding parameters, seeding policy, and retry policy in Appendix C.

Spend. The record reflects roughly $120 of Anthropic spend on the single-shot Haiku, Sonnet, and Opus runs (itemized in Appendix C) plus Azure quota for the non-Anthropic models; the Azure API returns no cost field, so this figure is not the total cost across all six models.

7.4 Results

We report four findings: incomparability (§7.4.1), the relevance mis-rank (§7.4.2), anti-monotonicity (§7.4.3), and the model-relative value of private context (§7.4.4). Table 1 consolidates the per-model verdicts; Tables 2, 4, and 5 give the underlying quantities.

7.4.1 Incomparability verified on all six models

On every model the empirical deficiency is mutually positive, $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ and $\widehat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$, and the uniform lower bounds $L_{\mathcal{D}}$ in both directions clear zero, verifying empirical incomparability at $\geq 90\%$ (Table 2). The crossover is stark: W_1 solves the API and trap cells and fails encoding, while W_2 solves encoding and fails the API and trap cells. For Haiku at $n = 40$ the carrying cells are

Table 1: Consolidated 6-model / 4-vendor verdict. All six models are run the same way, a single API call per attempt at $n = 40$ per cell (§7.3). "Anti-mono. verified" counts cells on which the interference upper bound clears $-\tau = -0.30$ at $\eta = 0.10$. "Argorder `none`" is the no-context baseline PASS on `api_argorder`, the model-relative-value spectrum. All figures are real measured runs; no modeled numbers.

Claim	Haiku	Sonnet	Opus	GPT-5.5	DeepSeek	Kimi-K2.6
Vendor	Anthropic	Anthropic	Anthropic	OpenAI	DeepSeek	Moonshot
n per cell	40	40	40	40	40	40
Incomp. verified ($\geq 90\%$)	✓	✓	✓	✓	✓	✓
$L_{\mathcal{D}}(W_1, W_2)$	+0.76	+0.72	+0.72	+0.76	+0.35	+0.68
$L_{\mathcal{D}}(W_2, W_1)$	+0.76	+0.76	+0.76	+0.76	+0.72	+0.76
Relevance mis-rank (trap)	✓	✓	✓	✓	✓	✓
Anti-mono. verified (# cells)	1	2	0 (repro.)	1	1	1
`api_argorder none`	2%	0%	40%	75%	2%	45%

Table 2: Verified incomparability: uniform $(1 - \eta)$ lower bounds $L_{\mathcal{D}}$ in both directions ($\eta = 0.10$). $L_{\mathcal{D}}(W_1, W_2) > 0$ and $L_{\mathcal{D}}(W_2, W_1) > 0$ together verify empirical incomparability. Both bounds are read off the same $2|\mathcal{D}|$ Clopper–Pearson intervals (Proposition 3).

Model	Vendor	n	$L_{\mathcal{D}}(W_1, W_2)$	$L_{\mathcal{D}}(W_2, W_1)$
Haiku-4.5	Anthropic	40	+0.762	+0.762
Sonnet-4.6	Anthropic	40	+0.715	+0.762
Opus-4.8	Anthropic	40	+0.715	+0.762
GPT-5.5	OpenAI	40	+0.762	+0.762
DeepSeek-V4-Pro	DeepSeek	40	+0.347	+0.715
Kimi-K2.6	Moonshot	40	+0.675	+0.762

`api_argorder` (`none` 2% / W_1 100% / W_2 0%) and `enc_amount` (`none` 0% / W_1 0% / W_2 100%), giving $L_{\mathcal{D}} = +0.762/+0.762$, far clear of zero.

Because incomparability is a two-directional claim, each row of Table 2 requires both bounds positive; the smaller of the two is the binding one. The two carrying cells are one API cell and one encoding cell, so the effective support of the surrogate is two cells. We therefore describe $\mathcal{D}$ as a fixed adversarial four-cell battery rather than a sampled task distribution: the verification is valid for what it covers, and what it covers is a single hand-built API-versus-encoding opposition replicated across six models.

Why Opus verifies incomparability but not interference. Opus verifies incomparability ($L_{\mathcal{D}} = +0.715/+0.762$) but not anti-monotonicity, and the reason is instructive. Incomparability is carried by stark cells whose proportions sit at 0 or 1, where the Clopper–Pearson interval is tight. The interference bound instead mixes the W_{1+}, W_1, W_2, and `none` arms: Opus's superset collapse on `api_post_ok` is real but milder ($100\% \to 38\%$, upper bound -0.23, short of $-\tau$), and its `api_argorder` baseline leaks (`none` = 40%; Remark 5), so neither cell verifies. Both facts are the same model-relative-value effect (§7.4.4): Opus partially already knows the convention, which shrinks the harm the verification can attribute to the added signal.

Table 3: The ranker spectrum on the probe tasks. Each cell is the ranker's top pick between $\{W_1, W_2\}$; "PASS" is the realized-utility ranking from the verified record. Dense and cross-encoder scores are deterministic; the LLM listwise row pools $n = 30$ judgments per task (unanimous, both presentation orders). $\times$ marks a mis-rank. The cross-encoder's `enc_amount` pick is a near-tie (-0.279 vs -0.305).

Ranker	Family	`trap` (PASS: W_1)	`api_post_ok` (PASS: W_1)	`enc_amount` (PA
bag-of-words cosine	lexical	W_2 $\times$	W_1	W_2
`bge-small-en-v1.5`	dense bi-encoder	W_2 $\times$	W_1	W_2
`e5-small-v2`	dense bi-encoder	W_2 $\times$	W_1	W_2
`ms-marco-MiniLM-L-6-v2`	cross-encoder	W_2 $\times$	W_1	W_1 $\times$
Haiku listwise (n=30)	LLM listwise	W_1	W_1	W_2

7.4.2 The relevance score mis-ranks the sources

On the trap, query-conditioned lexical relevance ranks $\varphi_{\text{lex}}(W_1) = 0.36 < \varphi_{\text{lex}}(W_2) = 0.44$, so it prefers W_2; yet realized PASS ranks $W_1 \gg W_2$ on all six models (Haiku 90% vs. 0%; Sonnet, Opus, GPT-5.5, and Kimi 100% vs. 0%; DeepSeek 98% vs. 0%). On the honest cells relevance *agrees* with PASS, on the API cells relevance ranks $W_1 > W_2$ and so does PASS, and on the encoding cell relevance ranks $W_2 > W_1$ and so does PASS, so the score fails *only* where utility diverges from topicality, exactly the regime Theorem 2 and Proposition 2 isolate. The `none` floor on the trap is 0% on all six models, and W_2 scores 0% there too, confirming both that the convention is unguessable and that encoding knowledge is useless on this task.

The ranker spectrum: dense retrievers and a cross-encoder fall for the trap; a listwise reranker escapes it. To test whether the mis-rank extends beyond bag-of-words, we scored $\{W_1, W_2\}$ against the probe tasks with representative open-weight rankers, two dense bi-encoders (`bge-small-en-v1.5`, `e5-small-v2`) and a cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`), and with a RankGPT-style LLM *listwise* reranker (Haiku single-shot, seeing the task and *both* sources; both presentation orders; $n = 30$ judgments per task). Table 3 reports the induced rankings. **Every feature-based ranker mis-ranks the trap**: both dense bi-encoders prefer W_2 on the trap while ranking both honest cells correctly, textbook topical behaviour, and the cross-encoder prefers W_2 on the trap with maximal confidence (logits -5.63 for W_1 vs $+3.24$ for W_2) while being near-tied (and narrowly wrong) on the honest encoding cell. The LLM listwise reranker, which can read the verifier-decisive coordinate rather than match surface features, tracks realized PASS everywhere (30/30 per task, both orders). This is exactly the boundary Proposition 2 draws: the impossibility hypothesis is *topicality*, invariance to the decisive coordinate, and the trap defeats the lexical, dense, and cross-encoder families that satisfy it while sparing the one ranker that reads content. The practical corollary is that escaping the topical class costs a model call per candidate, a probe of the same kind the deficiency verification is built from, not a free scalar.

7.4.3 Anti-monotonicity: a verified superset that hurts

The superset W_{1+}, which literally *contains* W_1's signal, sharply degrades the API cells relative to W_1 alone. The per-task interference $\Psi_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T) - \text{PASS}(W_2, T) + \text{PASS}(\texttt{none}, T)$ is verified negative ($\Psi_T \leq -\tau$, $\tau = 0.30$) on at least one cell for five of the six models (Table 4). Opus reproduces the effect (100% $\to$ 38%, $\Psi_T = -0.62$) but does not verify: the collapse

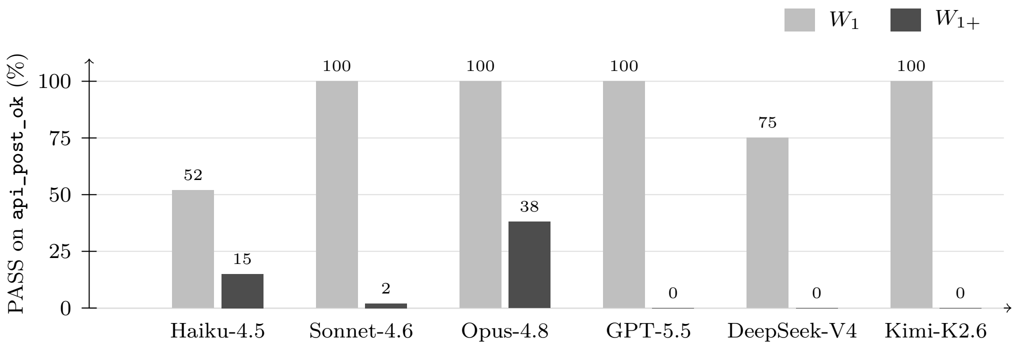

Figure 1: Anti-monotonicity across the capability ladder: PASS on the `api_post_ok` cell under the sufficient source W_1 (light) versus the strict superset $W_{1+} = W_2 +\!\!+ W_1$ (dark), per model (Table 5). The superset *contains* W_1's signal yet collapses performance on every model; the effect is large but not monotone in capability (Sonnet collapses hardest, Opus least). All runs single-shot, n=40. Haiku's W_1 does not reach ceiling on this cell single-shot (its verified collapse is on `api_argorder`, 100% $\to$ 5%; Table 4); Opus reproduces the collapse but does not verify.

is milder and its `api_argorder` baseline leaks, both of which we tie to the model-relative value of context in §7.4.4. This is a verified empirical falsification of Blackwell superset-dominance on live models (Theorem 3): strictly-more-informative context can hurt, and the deficiency framework detects it precisely where every monotone scalar must miss it. Figure 1 shows the collapse across the six models.

The empirical order inverts the idealized one. The idealized Blackwell order on $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ collapses the choice to the superset: W_{1+} reveals both coordinates, so $W_{1+} \succeq_B W_1$ and $W_{1+} \succeq_B W_2$, and "more signal is better" keeps $\{W_{1+}\}$ alone. The verified empirical relation for the fixed rule is the opposite shape: W_1 beats W_{1+} on the API cells, W_{1+} beats W_1 on the encoding cell, W_{1+} dominates W_2 (while W_1 and W_2 are the verified incomparable pair), so the correct empirical summary is the two-element Pareto front $\{W_1, W_{1+}\}$, an object no scalar top-1 rule can represent (Figure 2). This reads directly off the verified n=40 Haiku cells; the selection rule it implies, keep the verified non-dominated set rather than force the superset, is developed in a companion paper.

The leaky-baseline limitation is a stated protocol. The interference upper bound includes the term $+\mathrm{CP}_{\mathrm{hi}}(p_{\texttt{none}})$, so it can verify harm only on cells whose no-context baseline floors near zero. This is why anti-monotonicity verifies on `api_argorder` for Haiku (`none` = 2%, W_1 100%) and Sonnet (`none` = 0%) but not for GPT-5.5 (`none` = 75%), Kimi (45%), or Opus (40%), where a non-zero baseline mechanically shrinks Ψ_T's verifiable magnitude. We state this as a protocol up front (Remark 5): we verify anti-monotonicity only on cells whose `none` baseline floors near zero. Table 4 reports, per model, which cell carried the verification and its collapse, alongside the W_1 ceiling: DeepSeek's W_1 reaches only 75% on `api_post_ok` (an honest capability difference), so its $\Psi_T = -0.75$ starts from a lower point and is not directly magnitude-comparable to Sonnet's 100% $\to$ 2%.

Position is not the cause. The collapse is not a lost-in-the-middle or recency artifact. The length-matched padding control (Appendix B) puts filler in the same slot W_2 occupies inside W_{1+},

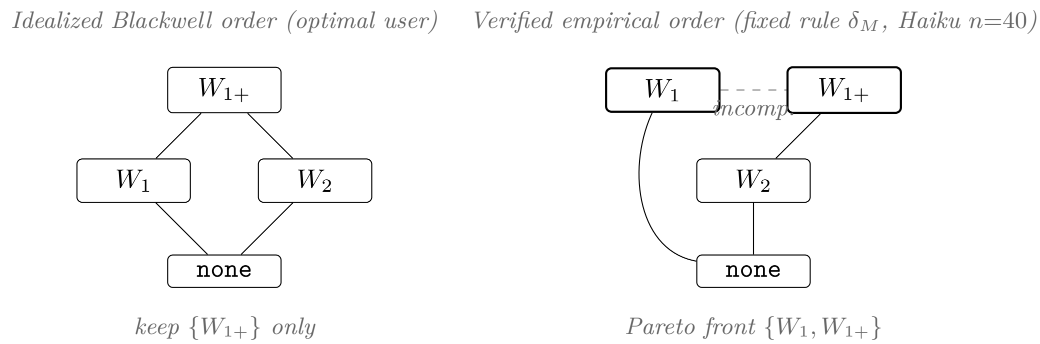

Figure 2: The paper's thesis in one picture, on the measured $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ with $W_{1+} = W_2 +\!+\, W_1$. **Left:** the idealized Blackwell order for the optimal user; W_{1+} reveals both coordinates, so "more signal is better" collapses the choice to the superset. **Right:** the verified empirical order for the fixed model rule (Haiku, n=40): W_1 beats W_{1+} on `api_argorder` (100% vs. 5%, anti-monotonicity, Table 4) while W_{1+} beats W_1 on the encoding cell, so the two are empirically incomparable (dashed) and the correct output is the two-element Pareto front, which no scalar top-1 rule can represent.

with W_1 last in both, and leaves W_1 at ceiling; holding position fixed and changing only the content is what turns a harmless arm into a harmful one, so the cause is the added signal, not where it sits.

Runtime fingerprint (a descriptive correlate). The harmful W_{1+} arm consistently emits more output tokens than W_1 across models. We report this as a reproducible correlate of the harmful arm, not a mechanism: more output on the failing arm could be effect rather than cause.

Robust, but not monotone in capability. The harm is large and robust across the Anthropic capability ladder but is *not* ordered by capability. On the two API cells the $W_1 \to W_{1+}$ collapse is Haiku 52%/100% $\to$ 15%/5%, Sonnet 100% $\to$ 2%/0%, and Opus 100% $\to$ 38%/15%: the middle model (Sonnet) shows the starkest collapse and the strongest (Opus) the mildest. An earlier two-point reading suggested the harm "intensifies with strength"; the third point falsifies that, and we claim only robustness across the ladder, not a capability trend.

7.4.4 The model-relative value of private context

The "private" conventions are genuinely unguessable for some vendors and partially known to others, and this varies continuously. The `none`-arm baseline on `api_argorder` forms a clean spectrum: Sonnet 0%, Haiku and DeepSeek 2%, Opus 40%, Kimi 45%, GPT-5.5 75% (Table 1; Figure 3). So the value of the private context, the model-relative quantity $I(S; W \mid \theta)$ of §3.3, varies continuously across models, and it tracks a vendor's training rather than raw capability: DeepSeek, a different vendor, has a tight prior like the smaller Claude models, while GPT-5.5's prior already holds much of the convention (`api_argorder` 75%, `enc_amount` 55%). Within Anthropic the ordering is not by capability either: Sonnet guesses least (0%) while the stronger Opus guesses more (40%). The verified per-model claims rest on cells unguessable *for the model under test* (e.g. GPT-5.5 `api_post_ok` `none` = 5%, trap = 0%).

Auditable per-cell record. The full per-arm PASS record for every carrying cell is Table 5 in Appendix B, so each verification can be reconstructed from the reported counts.

Table 4: Anti-monotonicity (interference) verification on the carrying cell per model. $\Psi_T = \mathrm{PASS}(W_{1+}) - \mathrm{PASS}(W_1) - \mathrm{PASS}(W_2) + \mathrm{PASS}(\texttt{none})$; "Upper" is the $\eta/4$ Clopper–Pearson upper bound $\overline{\Psi}_T$ of (7); the cell is verified harmful when Upper ≤ -0.30. The $W_1 \to W_{1+}$ column reports the PASS collapse. The verification clears only on cells whose `none` baseline floors near zero (the leaky-baseline limitation; Remark 5).

Model	Carrying cell	$W_1 \to W_{1+}$	Ψ_T	Upper	Verified
Haiku-4.5	`api_argorder`	100% → 5%	−0.93	−0.56	✓
Haiku-4.5	`api_post_ok`	52% → 15%	−0.38	+0.08	no (low ceiling)
Sonnet-4.6	`api_post_ok`	100% → 2%	−0.97	−0.64	✓
Sonnet-4.6	`api_argorder`	100% → 0%	−1.00	−0.69	✓
GPT-5.5	`api_post_ok`	100% → 0%	−0.95	−0.60	✓
DeepSeek-V4-Pro	`api_post_ok`	75% → 0%	−0.75	−0.36	✓
Kimi-K2.6	`api_post_ok`	100% → 0%	−1.00	−0.69	✓
Opus-4.8	`api_post_ok`	100% → 38%	−0.62	−0.23	no (milder)

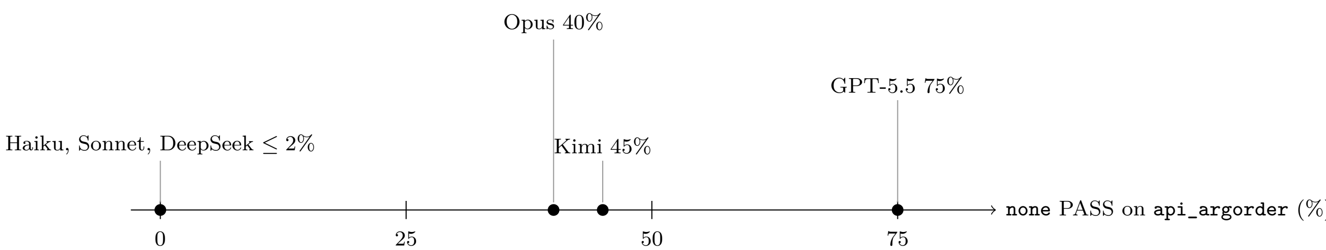

Figure 3: The model-relative value of private context as a spectrum: how often each model reproduces the private convention with *no* context supplied (`none` arm, `api_argorder`). The value of the same fixed source W_1 ranges from maximal (Sonnet, Haiku, DeepSeek: essentially unguessable) to largely pre-priced-in (GPT-5.5: 75% guessed), and the ordering tracks vendor training rather than capability, DeepSeek sits with the smaller Claude models despite being a different vendor, and Opus (the strongest Anthropic model) guesses more than Sonnet.

7.5 Generality and controls (summary)

Four follow-up experiments address three threat axes (source-pair generality, the context-length confound, and prompt-curability of the collapse); full protocols and numbers are in Appendix B. **(i) A second incomparable pair, different domain**: a private `cache`-module contract versus a key-normalization invariant is again verified incomparable on Haiku ($L_{\mathcal{D}} = +0.774/+0.496$), with one-directional dominance fully verified at $n = 80$; its interference is verified in sign ($\Psi_T < 0$, upper bound -0.21) but not at the $\tau = 0.30$ margin, because the 80% V_1 ceiling keeps the four-term bound wide. **(ii) Length-matched padding**: filler character-matched to W_{1+} leaves W_1 at ceiling (`api_argorder` 100% vs W_{1+} 10%; interference verified, upper -0.41), attributing the collapse to the added signal, not token count. **(iii) Routing instruction**: a one-line convention-routing note recovers most of the collapse (2% → 80%, 22% → 68%; both separated from W_{1+}) while remaining directionally below W_1 alone on the carrying cell; writing that note requires knowing which source is decisive for which task family, which is exactly the coordinate structure the partial order formalizes, so the cure presupposes the diagnosis. **(iv) Structured context**: wrapping the same two conventions in labeled XML sections *without* any routing statement mitigates only weakly (2% $\to$ 42% on the carrying cell, versus 80% with the routing note and 100% for W_1 alone; verified against the raw superset, still far below W_1), so segmentation alone does not cure the interference, the operative variable is routing knowledge. The mitigation hierarchy, raw superset < structure < routing < dropping the dominated signal, is itself a finding: it locates the failure in convention-to-task routing rather than in convention segmentation.

8 Discussion

With the evidence in hand, we step back to what it licenses. Our results recast a piece of practitioner folklore, "relevance does not transfer across tasks", as a structural fact about context sources. When two sources are Blackwell-incomparable, the order over them is partial, and any scalar score, by imposing a total order, must mis-rank them on some task. The empirical program does not merely illustrate this; it instantiates a single hand-built incomparable pair (a private API-call contract W_1 versus a private wire-encoding invariant W_2 on the same `ledger` module) and shows that the resulting double crossover, the relevance mis-rank, and an anti-monotonicity collapse all reproduce across six models and four vendors. We discuss what the evidence does and does not license, and where the formalization is a model rather than a proof of the realized object.

8.1 Scope of the claims

Three limits of scope govern everything above. **(i) Two registers.** The theorems are about the ideal user of a source; every number we measure is the pass rate of one fixed, non-optimal rule δ_M. Lemma 2 connects the two in the direction we need: a verified crossover on tasks where each source supplies its own decisive fact implies genuine Blackwell incomparability, and the reverse needs an extra assumption (Assumption 1) that our task design builds in and our measurements confirm. So we read the result as a statement about the fixed model on a fixed battery that *earns* the structural conclusion through the bridge rather than assuming it. **(ii) The estimator's name.** $\widehat{\delta}_{\mathcal{D}}$ does no minimization over garblings and ranges over a fixed set of tasks; it is a one-sided *value-deficiency surrogate*, not a Le Cam deficiency, and the two are not interchangeable for a fixed rule (Remark 4). What is rigorous is the verification, which we regard as the most robust part and which stands whether or not the deficiency analogy holds. **(iii) What the verification covers.** It is correct because it pays for picking the worst task with exact intervals and a union bound; its support is small (one API task and one encoding task carry the pair-1 result), each claim is controlled on its own at level η rather than all at once (§6.5), and we call $\mathcal{D}$ a fixed adversarial battery, not a task distribution.

8.2 Anti-monotonicity is real, verified, but not capability-ordered

The interference object $\Psi_T = \mathrm{PASS}(W_{1+}, T) - \mathrm{PASS}(W_1, T) - \mathrm{PASS}(W_2, T) + \mathrm{PASS}(\text{none}, T)$ measures whether appending the (irrelevant-to-this-task) W_2 to a sufficient W_1 harms performance. We verify $\Psi_T \leq -0.30$ on at least one cell for five of six models (Table 4), with collapses as stark as 100% $\to$ 0% ($\Psi_T = -1.00$, upper bound -0.69) on Sonnet and Kimi. This is a verified empirical falsification of superset-dominance for these fixed rules (Theorem 3): a literal superset of a sufficient signal can sharply reduce PASS, which no monotone-submodular scalar context-value score can represent.

Two corrections follow from the data. **(i) The harm is not monotone in capability.** An early two-point read (Haiku then Sonnet) suggested the collapse "intensifies with model strength"; the third Anthropic model (Opus) falsifies this. On `api_argorder` the collapse is 100 $\to$ 5% (Haiku), 100 $\to$ 0% (Sonnet), and 100 $\to$ 15% (Opus); the middle model (Sonnet) is the starkest and the strongest (Opus) the mildest. The defensible claim is robustness across the capability ladder, *not* a capability trend. **(ii) The interference verification is structurally cell-selected.** Its upper bound includes $+\mathrm{CP}_{\mathrm{hi}}(p_{\mathrm{none}})$, so it can only verify harm on cells whose none-baseline floors near zero. This is precisely the set of cells unguessable for the model under test, which couples the verifiable-harm set to the model-relative value of private context (§8.3). We treat this as a protocol (Remark 5): we verify anti-monotonicity only on floor-baseline cells, because a leaky baseline mechanically shrinks the verifiable Ψ_T. On `api_argorder` the harm is verified for Haiku (baseline 2%) and Sonnet (0%) but not for GPT-5.5 (baseline 75%), Kimi (45%), or Opus (40%).

Will scale restore Blackwell coherence? The capability non-monotonicity raises a question this framework makes precisely testable: is Blackwell coherence of the realized rule an emergent property of scale? Three points on one vendor's ladder cannot estimate that curve, and we decline to extrapolate from them; what the framework contributes is the *instrument*: coherence is now a verifiable property of any fixed rule, so "Ψ_T versus scale" is a measurable trajectory across model generations rather than a speculation, and the present data offer no comfort that the frontier is already there (GPT-5.5 verifies a 100% $\to$ 0% collapse; Opus reproduces the harm at strong point estimates). Two things would remain true under any scaling outcome. First, the theory treats δ_M as fixed *per measurement*, not as a claim that models are static: every new model or configuration is a new θ (§3.3), and every verified quantity is indexed by it, so model evolution is inside the framework, not outside it. Second, and decisively: anti-monotonicity is a coherence failure of the *rule* and could in principle vanish with scale, but incomparability is a structural property of the *sources*, mutually non-refining signals stay incomparable no matter how capable the reader becomes, so the paper's central claim, that selection is a partial-order problem, is invariant to how the scaling question resolves; scale can change only how gracefully a fixed rule degrades, never the order itself.

8.3 The value of private context is model-relative and tracks vendor training

The value of a private convention, $I(S; W \mid \theta)$, is itself a model-relative (θ-dependent) object, and the data give a clean continuous map of it. The none-arm baseline on `api_argorder`, how often a model reproduces the private convention with no context supplied, forms a spectrum: Haiku and Sonnet $\sim$0%, DeepSeek 2%, Opus 15%, Kimi 45%, GPT-5.5 75%. We do not claim this tracks capability. The decisive piece of evidence is DeepSeek: it has a tight, near-zero prior like Claude despite being a different vendor, while GPT-5.5's prior is wide (55%–75% guessing on `api_argorder/enc_amount`). A "stronger-model-knows-more" account predicts the spectrum to follow capability; it does not. We read $I(S; W \mid \theta)$ as tracking a vendor's training distribution rather than raw capability, which makes the theory's θ-dependence concrete and caps which cells verify per model.

8.4 Mechanism: the runtime fingerprint is a correlate, not an explanation

Every collapse is accompanied by output-token inflation on the harmful arm: W_{1+} emits between +87% and +281% more output tokens than W_1 on the API collapse cells (e.g. Haiku $+106\%/+87\%/+252\%$; Sonnet $+120\%/+131\%$; GPT-5.5 $+281\%/+274\%$). One might read this as "verbose thrashing" causing the failure, but the relationship is correlational and uncontrolled: the same superset arm also inflates output by +12% on the *still-passing* encoding cell, which undercuts a clean "verbosity causes collapse" account. We therefore treat the token fingerprint as a reproducible correlate of the harmful arm, not a mechanism.

Two points bound what this paper can and cannot say about mechanism. First, the claim structure is deliberately mechanism-agnostic: Theorem 3 verifies that the realized rule is not Blackwell-coherent *whatever* the internal cause, and the behavioral controls already constrain the hypothesis space from the outside, the harm is not position (order control), not token count (padding control), and is largely, though not fully, curable by explicit convention routing (instruction ablation), a signature consistent with convention interference or mis-binding rather than capacity exhaustion. Second, an activation-level account is genuinely out of reach for the objects studied here: four of the six models are closed-weight production APIs, and the phenomenon is cross-vendor, so any internal story must generalize across architectures we cannot open. The direct path to mechanism is to reproduce the collapse on an open-weight model with the same harness (the launchers speak any OpenAI-compatible endpoint) and then instrument attention and activation patterns on the contrasting arms; we regard this as the natural mechanistic companion study, downstream of, not prerequisite to, the verified behavioral claim.

Levels of description, and three readings of the phenomenon. The partial order is a decision-theoretic, behavioral-level frame; it does not compete with a computational account of how conditioning on added text changes the model's processing. The two are levels of description of the same event, and the verified behavioral facts are the constraints any computational account must satisfy. Three interpretive readings are compatible with the present data, and each is discriminable by a named experiment. **(i) A transient failure of current models**: discriminated by the coherence-versus-scale trajectory of §8.2, if Ψ_T shrinks toward zero across model generations, the anti-monotonicity half of the phenomenon was a passing artifact (the incomparability half, being a property of the sources, remains). **(ii) A property of fixed-compute inference**: a rule that spends the same forward computation regardless of context configuration cannot be optimal for every configuration, on this reading some coherence violation persists at every scale and only relocates. **(iii) Prompts as programs**: the added convention does not merely add information, it selects a different computation; this reading is supported by the routing result, a one-line control statement that adds no task-relevant information recovers most of the collapse, and by the structured-context ablation, labeled XML segmentation of the same content recovers only a minority of the collapse ($2\% \to 42\%$) while the routing statement recovers most of it ($\to 80\%$), locating the operative variable in convention-to-task *routing* rather than formatting or segmentation. We state these as interpretations, not findings; every verified claim in the paper is independent of which reading is true.

8.5 Limitations and threats to validity

We collect the scope restrictions here; each is stated once and sized to its weight.

From diagnosis to selection. The constructive selection rule the partial order implies, with its guarantees and evaluation against strong reranker baselines, is factored into a companion paper. Building the verified order on m candidates costs $m \cdot |\mathcal{D}| \cdot n$ verifier-gated calls, curation scale, not corpus scale; scaling verified selection to retrieval over 10^6 passages, by amortizing the probe into a learned deficiency predictor or running the partial-order step as a re-ranking head over a scalar shortlist, is the central open problem this work opens.

Source-pair generality. Incomparability and dominance hold for two hand-built pairs in two domains; the cross-vendor record is single-pair. Pair-2 interference is verified in sign ($\Psi_T < 0$ at $n = 80$) but not at the $\tau = 0.30$ margin, its 80% V_1 ceiling keeps the four-term bound wide; pair-2's relevance trap was an honest null (its prompt was not lexically V_2-leaning). A third domain and a vendor-general strong mis-rank remain inexpensive next steps.

Cross-vendor magnitudes are still only suggestive. All six models are now run the same way (single-shot, $n = 40$), so the run mode no longer confounds the comparison. What remains is the ordinary caution that different vendors' models differ in ways we do not control (training data, size, tokenizer), so the *magnitude* comparisons (the value spectrum, the Ψ_T ladder) should be read as suggestive; the binary within-model claims do not depend on any cross-vendor comparison.

The model is an idealization, twice. $\widehat{\delta}_{\mathcal{D}}$ verifies dominance and incomparability for a given model on a fixed battery, not the universal Blackwell order. And the projection construction models sources as clean projections onto independent coordinates; real chunks overlap, contradict, and leak (the measured θ-dependence is exactly such leakage), so the construction is the cleanest *sufficient* condition for incomparability, realized exactly by our hand-built pairs, and measuring verified-incomparability prevalence on natural document pairs is the direct next experiment.

Sampling and verification regime. The verification assumes n i.i.d. Bernoulli PASS draws per cell (Assumption 2), which holds because each draw is an independent single-shot call (protocol in Appendix C). Interference is verifiable only on cells whose `none` baseline floors near zero, stated as a protocol (Remark 5); marginal-versus-joint control is discussed in §8.1.

Ranker spectrum and synergy. The trap mis-rank spans lexical cosine and three representative open-weight learned rankers, with an LLM listwise reranker escaping as predicted, but the learned checkpoints are the small members of their families and the evidence is one trap in one domain. Positive interference (synergy) is untested by construction: W_1, W_2 are task-disjoint, so a complementary pair would be needed to test whether a superset can also *help* beyond its parts; we make no claim about it.

9 Conclusion

Context selection is, at bottom, a partial-order problem, not a scalar one. When two sources are incomparable, no scalar score can rank them correctly for all tasks (Theorem 2), which explains the field's "relevance does not transfer" folklore. We made the order measurable with a value-deficiency surrogate $\widehat{\delta}_{\mathcal{D}}$ and a distribution-free verification, and we ran the theory on six models across four vendors. On every model we verify ($\geq 90\%$) that two private sources are incomparable; on every model a real relevance score ranks them the wrong way; and on five of six we verify that adding a strictly larger source sharply lowers the pass rate (as deep as $100\% \to 0\%$), which no "more-is-better" scalar can predict. How much a private source is worth turns out to depend on the model: guessability runs from 0% to 75% across vendors and tracks a vendor's training rather than raw capability.

On scope: the theorems concern the idealized Bayes-optimal user of a source; the realized crossover, anti-monotonicity, and value-of-private-context are properties of *fixed* models measured directly, bridged to the structural claim by Lemma 2. The estimator is a regret-gap surrogate, not a Le Cam deficiency; the verification, which we regard as the most robust contribution, is correct and survives that re-labeling. All six models are run the same way (single-shot, $n = 40$), and the evidence spans two hand-built source pairs in two domains plus a length-matched padding control that attributes the anti-monotonicity collapse to signal content rather than added tokens; the cross-vendor record remains single-pair, cross-vendor magnitudes are only suggestive, and synergy is untested. The practical reading is modest and actionable: selection should be treated as partial-order-aware rather than scalar top-k, the right operational object is a verified deficiency surrogate rather than a relevance scalar, and "the best context" should be expected to be model-relative and only partially ordered. Stated at its sharpest, the impossibility is a theorem about *cheap* selection: per-source scalars (Theorem 2) and topical query-conditioned scores (Proposition 2), not all context selection. The measured spectrum realizes exactly that boundary: lexical cosine, two dense bi-encoders, and a cross-encoder all fall for the trap, while an LLM listwise reranker, which pays per-candidate computation to read the decisive content, escapes it. The hierarchy is computational: scalar scores cannot represent the order, and probes that read content can measure it, at probe cost. The remaining strengthenings, a vendor-general strong mis-rank, larger reranker checkpoints, and a third domain, are the natural next steps, and the experimental record indicates each is cheap to run.

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A Deferred proofs and technical material

Proof of Lemma 1

Proof. (a) For deterministic experiments a garbling G with $\kappa_{W_B} = G \circ \kappa_{W_A}$ must compute $\pi_B(S)$ from $\pi_A(S)$ alone (no further access to S). Such a measurable map exists iff π_B is $\sigma(\pi_A)$-measurable, i.e. $\mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A}$. Under coordinate independence and non-degeneracy, $\sigma(\pi_B) \subseteq \sigma(\pi_A)$ holds iff every coordinate in B is among those in A, i.e. $B \subseteq A$: if some $j \in B \setminus A$, then S_j is independent of $\pi_A(S)$ and non-constant, so it is not $\sigma(\pi_A)$-measurable, and no garbling can reconstruct it; conversely if $B \subseteq A$ the coordinate-deletion map is a (deterministic) garbling. *(b)* is the contrapositive of (a) applied in both directions. $\square$

Proof of Lemma 2

Proof. (a) Suppose, for contradiction, $W_2 \succeq_B W_1$. By Theorem 1, $V^\star(W_2, T) \geq V^\star(W_1, T)$ for every task T, in particular at T_a. But $V^\star(W_1, T_a) \geq \mathrm{PASS}_M(W_1, T_a) > g_a = V^\star(W_2, T_a)$, the first inequality because the optimal rule weakly improves on any fixed rule, the second by hypothesis, a contradiction. Hence $W_2 \not\succeq_B W_1$. Symmetrically, $\mathrm{PASS}_M(W_2, T_b) > g_b = V^\star(W_1, T_b)$ forbids $W_1 \succeq_B W_2$. Neither dominance direction holds, so W_1, W_2 are Blackwell-incomparable. The caps are essential: without them a PASS_M crossover is compatible even with Blackwell *equivalence*, since a fixed rule keyed to one signal's surface form can under-perform on the other, exactly the non-coherence phenomenon Theorem 3 verifies for (W_1, W_{1+}).

(b) Assumption 1 directly yields $\mathrm{PASS}_M(W_1, T_a) > \mathrm{PASS}_M(W_2, T_a)$ and the reverse at T_b, which is Definition 5's incomparability; the failure of the unconditional converse is witnessed by any fixed rule that violates monotonicity the optimal value obeys. $\square$

Proof of Proposition 1

Proof. Let G be the garbling with $\kappa_{W_2} = G \circ \kappa_{W_1}$. Any predictor $g \in \mathcal{V}$ acting on W_2's signal is realized on W_1's signal by the composite $g \circ G$, which lies in $\mathcal{V}$ by the closure hypothesis. Therefore the best $\mathcal{V}$-predictor under W_1 attains at least the predictive accuracy of the best under W_2, so its expected predictive risk is no larger; the risk-reduction relative to the common null signal is thus at least as large under W_1. The scalarity is immediate: $I_{\mathcal{V}}$ returns a single real number per source, so it induces a total order and, by Theorem 2, cannot rank a Blackwell-incomparable pair correctly for all tasks. $\square$

Proof of Proposition 3

Proof. Let E be the event that all $2|\mathcal{D}|$ Clopper–Pearson intervals contain their true proportions. Each interval fails with probability at most $\alpha = \eta/(2|\mathcal{D}|)$ (exact coverage of Clopper–Pearson [Clopper and Pearson, 1934]), so by the union bound $\Pr[E^c] \leq 2|\mathcal{D}| \cdot \alpha = \eta$, giving $\Pr[E] \geq 1 - \eta$. On E, for every T we have $p_{W_2,T} \geq L_{W_2,T}$ and $p_{W_1,T} \leq U_{W_1,T}$, so $p_{W_2,T} - p_{W_1,T} \geq L_{W_2,T} - U_{W_1,T}$; taking $\max_T$ on both sides, $\sup_T [p_{W_2,T} - p_{W_1,T}] \geq L_{\mathcal{D}}(W_1, W_2)$. Thus on E (probability $\geq 1 - \eta$) the true sup is at least $L_{\mathcal{D}}(W_1, W_2)$, which is the claimed lower confidence bound; if it is positive, the true sup is positive, verifying $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$. The reverse direction uses $L_{W_1,T}$ and $U_{W_2,T}$, which are endpoints of the *same* $2|\mathcal{D}|$ intervals already included in E; therefore both one-sided verifications are valid on the single event E, and their conjunction holds at $\geq 1 - \eta$ without a further Bonferroni factor. $\square$

Proof of Proposition 4

Proof. The four arms are independent Bernoulli samples (separate completions), so on the intersection of the four Clopper–Pearson events, probability $\geq 1-4\alpha = 1-\eta$ by the union bound, each true proportion lies in its interval. Substituting the coverage-respecting endpoint for each term of (6) (upper endpoints for the positively-signed $p_{W_{1+}}, p_{\texttt{none}}$, lower endpoints for the negatively-signed p_{W_1}, p_{W_2}) gives $\Psi_{T^\star} \leq \overline{\Psi}_{T^\star}$ on that event. For a post-hoc selected cell, replacing α by $\eta/(4|\mathcal{D}|)$ and union-bounding over the $4|\mathcal{D}|$ endpoints preserves $\geq 1-\eta$ simultaneous coverage. □

Proof of Theorem 3

Proof. $W_{1+} \succeq_B W_1$ by deletion of the extra coordinate (a deterministic garbling, Lemma 1(a)). By Theorem 1 any rule whose value respects $\succeq_B$ weakly improves under the dominating experiment. The hypothesized strict decrease $\mathrm{PASS}_M(W_{1+}, T^\star) < \mathrm{PASS}_M(W_1, T^\star)$ contradicts monotonicity, so δ_M is not Blackwell-coherent. A monotone-submodular scalar ϕ assigns $\phi(W_{1+}) \geq \phi(W_1)$ (monotonicity), hence orders W_{1+} above W_1 and cannot reproduce the realized reversal. □

Why a naive DKW bound fails

Remark 6 (Why a naive DKW bound fails). The Dvoretzky–Kiefer–Wolfowitz inequality [Dvoretzky et al., 1956] bounds the sup-deviation of *one* empirical CDF from *its own* population CDF over a single i.i.d. sample. Our object is different in two ways that DKW does not handle: it is a max over $|\mathcal{D}|$ *differences of two binomial means across distinct cells*, and the max *selects* the argmax-gap cell. A single-sample sup inequality neither spans distinct cells nor pays for the selection, so its nominal level under-covers a supremum over $|\mathcal{D}|$ comparisons; the Type-I error grows with $|\mathcal{D}|$. The multiplicity must be paid explicitly via the union bound (Proposition 3), not absorbed into a single-sample DKW envelope.

Relation to LLM-evaluation statistics

Remark 7 (Relation to LLM-evaluation statistics). The verification sits in a small but growing statistical-methodology literature for black-box model evaluation: paired-difference designs and clustered standard errors for evals [Miller, 2024], two-sample tests verifying behavioral differences between API-served models [Gao et al., 2025], and the recommendation to prefer exact frequentist intervals over CLT approximations when per-cell n is below a few hundred [Bowyer et al., 2025], which is precisely our regime ($n \in [20, 80]$ per cell) and precisely what the exact Clopper–Pearson construction provides.

Power and the role of n

The two verifications have different effective-n requirements, which is why at a common $n = 40$ Opus verifies incomparability but not interference. For a stark cell with empirical proportion at an endpoint ($\hat{p} \in \{0, 1\}$), the Clopper–Pearson interval collapses to a one-sided bound that clears 0 even at small n: the incomparability verification (5) is carried by such stark cells (W_1 at $\approx 100\%$, W_2 at 0% and vice versa), so it is easy to clear. The interference verification (7) mixes a possibly-intermediate `none` baseline and the W_{1+} arm; with non-endpoint proportions (as for Opus, whose collapse is milder and whose `api_argorder` baseline leaks) the four-term interval stays wide, so the same n that verifies incomparability need not verify interference.

Table 5: Per-cell PASS rates (%) for all four verified cells, by arm, all single-shot at $n = 40$. Every model was measured on every cell and arm. The encoding rows (`enc_amount`) carry the $W_2 > W_1$ direction of incomparability: on every model W_2 solves encoding while W_1 fails it, mirroring the API cells where W_1 solves and W_2 fails.

Model	Cell	`none`	W_1	W_2	W_{1+}
Haiku-4.5	`api_post_ok`	0	52	0	15
Haiku-4.5	`api_argorder`	2	100	0	5
Haiku-4.5	`enc_amount`	0	0	100	100
Haiku-4.5	`trap_store_wire`	0	90	0	92
Sonnet-4.6	`api_post_ok`	0	100	0	2
Sonnet-4.6	`api_argorder`	0	100	0	0
Sonnet-4.6	`enc_amount`	0	0	98	80
Sonnet-4.6	`trap_store_wire`	0	100	0	65
Opus-4.8	`api_post_ok`	0	100	0	38
Opus-4.8	`api_argorder`	40	100	0	15
Opus-4.8	`enc_amount`	0	0	98	85
Opus-4.8	`trap_store_wire`	0	100	0	100
GPT-5.5	`api_post_ok`	5	100	0	0
GPT-5.5	`api_argorder`	75	100	0	0
GPT-5.5	`enc_amount`	55	0	100	100
GPT-5.5	`trap_store_wire`	0	100	0	100
DeepSeek-V4-Pro	`api_post_ok`	0	75	0	0
DeepSeek-V4-Pro	`api_argorder`	2	68	0	5
DeepSeek-V4-Pro	`enc_amount`	0	0	68	65
DeepSeek-V4-Pro	`trap_store_wire`	0	98	0	98
Kimi-K2.6	`api_post_ok`	0	100	0	0
Kimi-K2.6	`api_argorder`	45	100	0	0
Kimi-K2.6	`enc_amount`	0	0	95	92
Kimi-K2.6	`trap_store_wire`	0	100	0	100

B Extended experimental record

Generality and controls: full protocols

Four follow-up experiments address three threat axes: source-pair generality, the context-length confound, and whether the anti-monotonicity collapse is curable by cheap prompting.

A second incomparable pair, different domain. We repeated the protocol with a second hand-built pair on a private `cache` module: V_1, the `cache.put(key, value, *, ttl)` call contract (argument order, required keyword-only `ttl`, a True/False "newly inserted" flag, a `CacheError` type), versus V_2, a key-normalization invariant (a fixed `kx7-` namespace prefix and an underscore rule). On Haiku at $n = 40$ the pair is again verified incomparable ($L_{\mathcal{D}} = +0.774/+0.496$, both directions clear of zero) with a clean double crossover (V_1 solves the contract and trap cells, V_2 solves key-normalization, all none-baselines floor at 0%), and the dominance control again verifies one-directionally. Anti-monotonicity reproduces in direction at $n = 40$ (the superset collapses the contract cell 80% → 28%, $\Psi_T = -0.53$); re-running the four arms of the contract cell at $n = 80$

shows the collapse is stable ($80\% \to 29\%$, $\Psi_T = -0.51$) with Clopper–Pearson upper bound -0.21, which verifies *strictly negative* interference ($\Psi_T < 0$ at $\geq 90\%$) but does not clear the pre-registered magnitude margin $\tau = 0.30$: the binding constraint is the V_1 ceiling itself (80%, versus pair-1's $\sim$100%), which keeps the four-term bound wide regardless of n. We report pair-2 interference as a verified sign, not a verified magnitude. The $n = 80$ run also strengthens the dominance control to a fully verified one-directional dominance ($U_{\mathcal{D}} = 0.000$ with a verified strict win). The relevance trap for this pair was a null, its prompt vocabulary leaned toward V_1 ($\varphi(V_1) = 0.39 > \varphi(V_2) = 0.25$), so relevance *agreed* with PASS, so the strong mis-rank remains a pair-1 result. The contribution of the second pair is therefore *generality of the incomparability and dominance results to a new domain*, not a second mis-rank.

A length-matched padding control. Anti-monotonicity compares W_1 against the strictly longer W_{1+}, and added input length alone is known to degrade performance [Du et al., 2025], so the collapse could in principle be a token-count effect rather than a signal effect. To isolate this we added a padding arm W_{pad}: filler text in the same "team context (private)" register as W_2, placed in the same position W_2 occupies in W_{1+} (padding first, W_1 last), character-matched to W_{1+} within 13 characters, but carrying no coordinate any verifier checks (office-logistics content). On single-shot Haiku at $n = 40$ on the two collapse cells, padding is free while the equal-length dominated signal is catastrophic: on `api_argorder`, W_1 98%, W_{pad} 100%, W_{1+} 10%; on `api_post_ok`, W_1 50%, W_{pad} 65%, W_{1+} 10%; and the same-run interference verification clears on `api_argorder` ($\Psi_T = -0.82$, upper bound -0.41). The anti-monotonicity harm is therefore attributable to W_2's signal content interacting with the fixed rule, not to added tokens. One residual axis is not isolated by this control: the padding is off-topic for the `ledger` module while W_2 is on-topic, so "topical but verifier-irrelevant" text remains a smaller, untested middle case.

A routing-instruction ablation: the cure presupposes the diagnosis. If the collapse reflects convention confusion, an explicit routing note might prevent it, and if a one-line prompt fully cured the harm, its practical significance would shrink. We measured this directly: the arm W_{1+}^{instr} appends to W_{1+} a single routing sentence stating which convention serves which task family ("for tasks that call the ledger API use only the API call contract; the wire-encoding invariant applies only to encode/decode tasks"). On single-shot Haiku at $n = 40$ the instruction recovers most, but not all, of the collapse: on `api_argorder`, $2\% \to 80\%$ (W_1 alone: 100%), and on `api_post_ok`, $22\% \to 68\%$ (W_1 alone: 50%); both recoveries over W_{1+} are large (non-overlapping exact 95% intervals), while the residual gap below W_1 alone on the carrying cell (80% vs. 100%, intervals $[0.62, 0.92]$ vs. $[0.90, 1.00]$) is directional but not separated at this n. We pre-commit to the two readings jointly. First, the harm is substantially mediated by convention interference that explicit routing knowledge prevents, so pipelines that can attach per-task routing labels have a cheap mitigation. Second, and decisively for the paper's thesis: *writing the routing sentence requires already knowing which source is decisive for which task family*, which is precisely the coordinate structure the partial order formalizes and the deficiency verification measures; the cure is an injection of selection-time coordinate knowledge into the prompt, not a scalar score, and no query-independent scalar can supply it (Theorem 2). Formally, δ_M with the routing note is a *different, more Blackwell-coherent rule*; the verified falsification of Theorem 3 concerns the base rule and is unaffected, and even under the instructed rule, keeping the dominated signal is at best equal and directionally worse than dropping it, so the selection-level conclusion, keep the non-dominated set rather than forcing the superset, survives the ablation intact.

A structured-context ablation: segmentation is not the bottleneck. To separate "the model cannot segment the two conventions" from "the model cannot route them to task families," the arm W_{1+}^{xml} wraps the same two conventions, in the same order as W_{1+}, in labeled XML sections (<wire_encoding_invariant>, <api_call_contract>) with no routing statement. On single-shot Haiku at $n = 40$ per cell, structure alone mitigates weakly: `api_argorder` 42% (vs. W_{1+} 2%, W_{1+}^{instr} 80%, W_1 100%) and `api_post_ok` 25% (vs. 12%, 68%, 57%). The structure-only mitigation is separated from the raw superset on the carrying cell (exact 95% intervals $[0.27, 0.59]$ vs. $[0.001, 0.13]$) but remains far below W_1 alone; on `api_post_ok` it is not separated from the raw superset. Protocol note: the W_{1+}^{xml} arm was measured in a clean arm-only re-run (its arm in the first run was invalidated by a transport-failure window and scored no completions; the re-run had zero transport errors), so its comparison against same-day baseline arms is across runs rather than within one run. The resulting mitigation hierarchy, raw superset < structure < routing < dropping the dominated signal, locates the failure in convention-to-task routing.

C Reproducibility and protocol

All reported numbers come from real model calls whose outputs are checked by executable verifiers; there are no modeled or aspirational figures. The complete artifact bundle (the measurement harness, the executable verifiers, and every per-model run log and result JSON cited below) is available at https://github.com/abilous-ti/blackwell-llm-context. The harness (`tokenbench/measure_blackwell.py`) is Python-standard-library only (no third-party dependencies; the Clopper–Pearson endpoints use a stdlib regularized incomplete beta), is cross-OS, and implements the surrogate $\widehat{\delta}_{\mathcal{D}}$, the uniform incomparability verification (`certify_incomparable`, `clopper_pearson`), the dominance control (`certify_dominance`), the interference verification, the sample-size helpers (`required_n`, `required_n_stark`), and the lexical-relevance scalar (`lexical_relevance`). It includes launchers for the Anthropic CLI (`claude -p`), the Azure Responses API, and OpenAI-compatible chat completions; the launcher auto-detects API style from the endpoint, and all API keys are read from the environment (never written to a file).

Protocol. The verifier stubs a fake `ledger` module that accepts *only* the exact private contract (it rejects wrong positional arg order, a missing `memo=` keyword, and the wrong exception type); return code 0 is PASS. The four verified cells are `api_post_ok`, `api_argorder` (`memo` arrives via a parameter, forcing the keyword-only fact), `enc_amount`, and `trap_store_wire`; the arms are $\{\text{none}, W_1, W_2, W_{1+}\}$, with W_{pad} (length-matched neutral filler $+\!+\, W_1$) added for the padding control. All six models are queried in a single-shot completion (one call, no tools, code extracted from the returned text), and PASS is measured over $n = 40$ i.i.d. completions per cell at $\eta = 0.10$.

Decoding, seeding, and retries. All runs use the provider's default decoding configuration: the harness sets no explicit temperature, top-p, or seed (the Claude CLI and the Azure/OpenAI-compatible launchers are invoked without sampling overrides), so the n draws per cell are independent samples from each provider's default decoder, each a fresh single-turn call with a 200-second timeout. Transport-level failures (empty or zero-token results) are retried up to three times with quadratic backoff; a draw that still fails is scored as a failure, never as a pass. Where a whole arm was hit by a transport-outage window, we re-ran that arm in isolation (the Opus W_{1+} arm; noted in the artifact list), so no reported cell mixes outage-scored draws with real ones.

Artifacts. Per-model run logs and result JSONs are in the artifact repository (https://github.com/abilous-ti/blackwell-llm-context) under the pattern `blackwell_*.{log,json}`: the six single-shot $n = 40$ model runs (`blackwell_haiku_ss_n40`, `blackwell_sonnet_ss_n40`, `blackwell_opus_ss_n40` with its arm-only re-run `blackwell_opus_ss_W1plus`, `blackwell_gpt55_n40`, `blackwell_deepseek_n40`, `blackwell_kimi_n40`), and the follow-up controls `blackwell_pad_n40b` (length-matched padding, 400 calls), `blackwell_pair2_n80` (second source pair), `blackwell_instr_n40` (routing-instruction ablation), `blackwell_xml_n40` / `blackwell_xml_arm_only` (structured-context ablation), `reranker_trap_n30` (LLM listwise reranker probe; `tokenbench/measure_reranker_trap.py`), and `dense_trap.json` (dense/cross-encoder probe, deterministic local scoring of open-weight checkpoints at zero API cost; `tokenbench/measure_dense_trap.py`). The full per-cell record is in `EXPERIMENT-BLACKWELL.md` and the theory hardening (formalization and proof sketches) in `BLACKWELL.md`.

Compute and cost. Total measured Anthropic spend was approximately $130: the single-shot Haiku, Sonnet, and Opus runs (Opus the costliest, plus its arm-only re-run) and the follow-up controls (padding, second pair, routing, structured-context). The non-Anthropic runs (GPT-5.5, DeepSeek, Kimi) used Azure quota; the Azure API returns no cost field, so this figure is the Anthropic spend, not the total across all six models. The dense/cross-encoder probe ran locally on open weights at zero API cost.

Reference verification. The bibliography was assembled in part with automated literature search and every entry was subsequently verified against arXiv/DBLP/publisher records; all entries resolve to real papers, and the classics (Blackwell 1953; Le Cam 1964/1986; Torgersen 1991; Ethayarajh et al. 2022) were checked against the primary sources.